

Compiling the Fly Brain

What a Complete Wiring Diagram Does and Does Not Determine

The fly-brain project

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Introduction: What Does a Wiring Diagram Actually Compute?

The question

A few years ago, “the complete connectome of a fly” was an aspiration. Today it is a file. The male *Drosophila* central nervous system, as reconstructed by Janelia FlyEM and Google Research, is 165,122 neurons and 104 million synapses, and the whole graph loads into memory on a laptop in a few seconds. So the obvious next question is: now that we have it, what does it *compute*?

I want to be precise about what I mean by this, because there is a weak version of the question and a strong version, and most of the interesting content is in the difference.

The weak version is: given recordings of neurons, what do they correlate with? This is a good question and neuroscience has excellent tools for it. But it is not the question a wiring diagram lets you ask for the first time.

The strong version is: given *only* the wiring, with no recordings at all, can you recover the algorithm? Can you say which cell populations form a continuous attractor, which ones rotate a vector, which ones implement an associative memory? And can you say it in the only form that can actually be tested, namely as a prediction of what happens when you silence a specific named cell type?

This book is about a project that tries to answer the strong version, and about what it found. The short summary of what it found is: the wiring gets you surprisingly far, it gets you less far than the wiring itself seems to promise, and the size of that gap is one of the most useful numbers you can compute.

Thinking of it as a compiler

The mental model the project uses is that of a compiler, and I think the analogy is worth taking seriously rather than treating as a slogan.

A compiler does not understand a program by reading it once. It lowers the source into an intermediate representation, runs a series of analysis passes over that representation, and emits artifacts that can be executed and checked. Each stage is narrow and testable on its own. The pipeline here does the same thing to a nervous system:

biology

- > canonical neural IR
- > motifs and structural signatures
- > candidate operators

- > ensembles of executable models
- > ranked discriminating experiments
- > validated models and reduced algorithms
- > engineering benchmarks

The connectome is lowered to a species-agnostic intermediate representation (the same loader also reads the worm). Analysis passes detect candidate *operators*: the computational primitives the wiring appears to implement. Each operator becomes not a model but a *family* of executable models, one per setting of the parameters the wiring does not fix. And the disagreements inside that family become the product: a ranked list of experiments that would tell the family members apart.

The reason the analogy matters is that it tells you where to be suspicious. A compiler pass that silently assumes something about the source is a bug. Likewise, a dynamics stage that silently re-imports an assumption about the structure is a bug, and a lot of the discipline in this book is about catching that.

Two very different ways to run the same graph

The project actually runs the connectome in two ways, and they fail in different places, which is the point.

The first way is the compiler proper. You analyse the graph structurally, you build small circuit-level model ensembles (tens of models, each with a few hundred to a few thousand cells), and for one circuit you also build a continuous field model. This is Parts I to III. It produces sharp claims about specific circuits and specific experiments.

The second way is to run the whole thing. All 165,122 neurons as conductance-based integrate-and-fire units, inside a physics-simulated body with a compound eye, taste, touch, smell and heat, in a web browser. This is Chapter 14. It produces a different kind of evidence: which behaviours the raw wiring supports once it is embodied, and which behaviours had to be supplied by code that lives outside the graph.

If you only did the first, you could fool yourself into thinking that a circuit-level success generalises. If you only did the second, you could not tell *why* anything works or fails. Doing both is more work but it is the only way to get an honest picture.

Why start with the heading circuit

The first circuit worked end-to-end is the fly's heading system in the central complex: the ring attractor formed by the EPG, Delta7, PEG and PEN neurons. The reason is simple. It is the best-understood circuit in any complete connectome. Physiologists already know it holds a bump of activity that encodes heading, they have imaged it, they have perturbed it, and they have written down what happened.

That makes it a calibration instrument. The test is:

Can the machinery recover a known computation from structure plus limited functional evidence, without the answer being written into the detection rules?

If it fails here, nothing else it says can be trusted. If it succeeds *and* can say where it is uncertain, then the same machinery can be pointed at circuits nobody understands yet, and its uncertainty statements there mean something.

What a good answer looks like

Let me be explicit about what the pipeline is designed to output, because it is not a simulation and it is not a single model. It is a statement like this one:

Across the parameterizations the wiring permits, this circuit supports a low-dimensional heading state, but only in a narrow gain regime. Three mechanisms remain compatible with baseline activity. Silencing this cell type distinguishes them.

To get there, the project tracks three different kinds of agreement between models and refuses to conflate them:

1. Task-performance agreement. The models solve the same problem.
2. Neural-response agreement. The models' population activity matches.
3. Perturbation-prediction agreement. The models break the same way.

Models very often agree on (1) while disagreeing completely on (3). That asymmetry is not a nuisance, it is the resource the whole method runs on: find the perturbation where the surviving hypotheses diverge the most, rank it, and go check it against real data.

How the book is organised

Part I is about the data and what you can learn from the graph before running any dynamics. Chapter 2 covers the connectome as a data structure and the intermediate representation. Chapter 3 covers the generic structural passes: flow, sign, motifs, spectrum, and a cross-species comparison against the worm. Chapter 4 walks through the circuits region by region.

Part II is the method. Chapter 5 presents the operator catalogue, twelve detected computational primitives with their evidence and their counterevidence. Chapter 6 explains why a connectome constrains a family of models rather than a model, and how a family gets turned into a ranked experiment list.

Part III works three circuits in detail. Chapters 7 to 9 cover the heading circuit as a spiking ensemble, as a continuous field, and then the argument between the two, which published data settled. Chapter 10 covers the PFN to $h\Delta B$ vector shift, where the wiring overstates. Chapter 11 covers the mushroom body, where the method returns a negative.

Part IV assesses. Chapter 12 benchmarks the decompiled heading estimator against standard filters, with the baselines given fair parameters. Chapter 13 describes the compiler machinery that turns an operator into an executed ensemble by writing a spec instead of a script. Chapter 14 reports what happened when the whole connectome was run in a body. Chapter 15 pulls the results together and lists what the connectome provably cannot tell you.

Three rules I would ask you to hold the book to

1. Negative results are recorded, not tuned away. When a mechanism fails under noise, or a circuit does not do what the wiring suggests, that is reported as a finding and left in the artifact.
2. Baselines are matched. A biological estimator is never declared better than a Kalman filter that was handicapped. Chapter 12 goes out of its way on this.
3. Disagreement between formalisms is data. When the spiking model and the field model disagree, the disagreement is written into the output file with a flag, and then resolved against biology, not

smoothed over.

If you find a place where the book breaks one of these, that is a bug in the book.

The Data, and the Intermediate Representation

What the object actually is

Before talking about what the connectome computes let me be very concrete about what it *is*, because a lot of the later conclusions come down to properties of the data structure.

The analysis runs on the male *Drosophila* CNS connectome, version 1.0. As a graph it is:

quantity	value
neurons	165,122
connections (3 or more synapses)	10,511,038
synapses	104,213,652
cell types	11,752
per-connection synapse count range	3 to 2,591

Each neuron carries a type label, a hemisphere, a superclass, a predicted neurotransmitter, and a volume. Each connection carries a synapse count. That is it. Notice how little annotation this is relative to the raw data: the neurotransmitter predictions come from a 2.7 GB table of per-synapse probabilities, and the neuron volumes come from a 4.3 GB neuron table that is otherwise unused. Only a handful of scalars per neuron turned out to be load-bearing.

Three properties of the data drive everything downstream.

Edges carry synapse counts, not strengths. A weight of 25 means “25 synaptic contacts were reconstructed between these two cells.” It does not mean “this connection moves the postsynaptic potential by 25 of anything.” The mapping from count to conductance is an assumption, and it turns out to be the single most consequential assumption in the whole pipeline. Chapter 11 is essentially a case study of it failing informatively. There is also a nice information-theoretic hint about this: when the graph is compressed for the browser, the synapse counts cost 4.04 bits each on their own and 3.95 bits with the best context model found. In other words, knowing everything else about the graph tells you almost nothing about a connection’s count. Counts carry information the rest of the structure does not predict.

Signs are predictions, and some are graded. Each synapse’s effect sign comes from the presynaptic cell’s predicted transmitter: acetylcholine excitatory, GABA and glutamate inhibitory, histamine inhibitory, monoamines treated as modulatory. For 3,602 neurons there is no consensus call, and rather than guess, their sign is taken as a graded value, $P(\text{ACh plus monoamines}) - P(\text{GABA plus glutamate})$.

plus histamine), from the per-synapse predictions. Only 0.5% of synaptic weight ends up unsigned. A cell whose transmitter is unknown contributes nothing in the dynamical models. This sounds like a small technical choice and it is not: an early version that treated unknown as excitatory spread odour activity to every glomerulus in the antennal lobe.

Types are fine-grained and structured. Labels like KCg-m, MBON01 to MBON35, PPL103, PEN_a, EPG let you select populations by pattern. Instance names go further: EPG(PB08)_L4 encodes a protocerebral-bridge glomerulus and a side, hDeltaB_08_C7 encodes a fan-shaped-body column, ORN_DA1 encodes a glomerulus. The central-complex analysis in Chapters 7 to 10 is only possible because these positional labels exist. Keep this in mind when thinking about other datasets: the compiler needs geometry, and here the geometry arrives through the naming scheme.

What the graph leaves out

Just as important is what is *not* in the file:

- Gap junctions. Electrical synapses are invisible in this EM reconstruction. The one that turned out to matter for behaviour, giant fibre to the jump motor neuron TTMn, had to be added by hand in the whole-brain model (Chapter 14).
- Connections under 3 synapses. They were dropped when the graph was packed. The motif census uses 3 per target neuron. The whole-brain model, after calibration, uses connections of 6 or more.
- Anything about strength, time constants, adaptation, or neuromodulatory state. Chapter 6 enumerates these because they become the axes of the ensembles.
- Locality is real, though. When the neurons were renumbered by cell type and then by the 3D position of their skeleton centroid, the entropy of “which neuron does this one connect to” fell from 11.8 bits to 8.2 bits per connection, and 28% of a neuron’s targets also appeared in the target list of the neuron numbered just before it. Type and position predict roughly 3.6 bits of each connection. That is the stereotypy that makes the type-level analyses in Chapter 3 meaningful.

The intermediate representation

Everything that follows runs on a species-agnostic IR rather than on the fly’s native tables. The IR is deliberately small:

field	meaning
names, types, scn, cln	cell name, cell type (bilateral homologues share one), superclass, class
side	1 left, 2 right, 3 midline, 0 unknown
sign	+1 excitatory, −1 inhibitory, 0 unknown or modulatory
sensory, motor	pins for the flow-depth computation (sensors 0, outputs 1)
indptr, indices, weights	the chemical graph in CSR form, weight = synapse count
gap_indptr, gap_indices, gap_weights	the electrical graph, symmetric, empty where absent

field	meaning
meta	provenance, sign convention, per-dataset thresholds

Two loaders exist. `load_fly()` reads the packed male-CNS tables and was checked to reproduce every generic number in the fly’s own structural report exactly. `load_worm()` reads the Cook et al. 2019 adult hermaphrodite *C. elegans* connectome from the Netzscheuler CSV dumps, assigns signs from the OpenWorm cell table (acetylcholine +1, GABA and glutamate −1, monoamines 0, covering 179 of 302 neurons, with the unassigned remainder almost entirely pharyngeal), and folds bilateral pairs and serial numbers into classes: AVAL/AVAR become AVA, DA01 to DA09 become DA, giving 169 types over 454 cells. The worm’s gap junctions are kept as a second, symmetric graph, which is a real edge type the fly IR lacks.

Why bother with an IR at all? Because the analysis passes in Chapter 3 (flow depth, motifs, sign structure, spectral properties) should not know which species produced the graph. If they did, a cross-species comparison would be rhetoric. With the IR it is a measurement.

What the worm tells you about the fly

Running both animals through the same passes gives this table:

measure	worm (Cook 2019)	fly (male CNS)
cells / connections / synapses	454 / 4,879 / 28,113	165,122 / 10.5 M / 104 M
forward / feedback / lateral	12% / 72% / 16%	37% / 12% / 50%
weight		
giant strongly connected component	61% of cells	97% of cells
reciprocal weight	29%	27%
dominant neuron-level two-cycle	I↔I (371 pairs)	E↔I (616,156 pairs)
midline-crossing weight	37%	22%
feedforward-inhibition share of strong E edges	17% (z 3.7)	80% (z 297)
spectral radius of the signed weight matrix	136	3,771
dominant-eigenvector participation	10.8 cells	49.9 cells
where the dominant mode lives	RMD, RIA, SMD (head steering)	ILN1/ILN2 antennal-lobe local neurons
hub tail exponent, in / out	2.6 / 1.9	1.0 / 1.15
unsigned weight	62%	0.5%

Some of what this says:

1. The worm is a feedback machine and the fly is a layered one. 72% of the worm's synaptic weight runs backward in the depth ordering, against 12% in the fly. Half the fly's weight is lateral, meaning between neurons at the same depth. These are genuinely different graph regimes, not the same regime at different scale.
2. The characteristic loop is different in kind. The fly's dominant two-cycle is feedback inhibition, a cell that excites another and is inhibited back. The worm's is mutual inhibition, which is the command-interneuron rivalry that worm locomotion is known for. At type level the worm's mutual-inhibition enrichment washes out (z about -0.4) because worm motor classes are single cells, so the motif only exists at neuron resolution. That is itself a lesson: the right resolution for a motif census depends on the animal.
3. The dominant dynamical mode identifies itself. On the worm, the leading eigenvector of the signed weight matrix lands on RMD, RMDV, RIA and SMD, the head-steering circuit that oscillates during foraging. Nobody told the analysis that. The same measurement on the fly lands on the lateral antennal-lobe local neurons. I find this the most encouraging single result of the cross-species pass, because it is a "found, not annotated" outcome.
4. The worm's sign structure is mostly unknown. 62% of its weight is unsigned against 0.5% for the fly, so any sign-conditioned conclusion about the worm is weaker by construction. The null model in Chapter 3 swaps about 100 edges per rewire on a 330-edge type graph, so the worm's z -scores should be read as directional.

The comparison does double duty. Scientifically it is interesting. Methodologically it is the check that the passes are actually species-agnostic rather than fly-shaped code that happens to run on a worm.

Where the IR stops

The IR answers "what is wired to what, how strongly, with what sign." It cannot answer "what does it do." That needs dynamics, which is Chapter 6 onward. The boundary is deliberate and it is the book's core methodological commitment: structure first, then dynamics, and never let the dynamics stage quietly re-import a structural assumption.

A small example of what that discipline looks like in practice. The transmitter table maps each synapse's sign from the presynaptic cell, and unknown yields zero effect rather than a guessed sign. When an early diagnostic found the mushroom-body interneuron APL reading `undefined` for its transmitter, the right response, and the one taken (Chapter 11), was to trace the sign path and find out, not to patch APL to inhibitory because "everyone knows APL is GABAergic." It is GABAergic. But the check had to be made, because a sign bug would have produced the same null result invisibly.

Structure Before Dynamics: What the Graph Alone Tells You

Before any neuron fires in any model, the wiring diagram already contains a lot of computation-shaped information. This chapter is about what the generic passes find, and just as importantly, about where structure stops being enough. Everything here is obtained without simulating a single spike.

The brain is shallow going forward and deep going around

Do a breadth-first search from every sensory neuron over connections of 5 or more synapses, and count how many hops each neuron is from the nearest sensor:

hops from sensory	0	1	2	3	4	5+	unreachable
neurons	15,912	27,495	84,711	35,214	902	31	857

Half the descending neurons are one hop from a sensory neuron and nearly all the rest are two. 465 of 708 motor neurons are one hop. So the longest sensor-to-muscle chain the wiring *needs* is three synapses, and almost every neuron in the brain is within three hops of a sensor.

A more careful ordering uses a harmonic depth: pin sensory neurons at 0 and motor and efferent neurons at 1, and give every other neuron the mean depth of its partners. The superclasses then fall into a sensible order: sensory 0, antennal-lobe local neurons 0.09, projection neurons 0.15, Kenyon cells 0.19, MBONs 0.26, central-brain intrinsic 0.33, central complex 0.35, descending 0.49, VNC intrinsic 0.52, motor 1. Measured against this order, synaptic weight splits as:

- 50% lateral, between neurons at the same depth.
- 37% forward, in the sensory-to-motor direction.
- 12% feedback.

The optic lobe alone is 89,390 neurons of lateral processing. So the picture is a feedforward skeleton three synapses deep, with most of the wiring, and therefore most of the computation, happening sideways and backwards. If you want to interpret anything else in this book, this is the fact to keep in mind: whatever long processing the fly does, it does by recurrence, not by depth.

One giant loop. 97.2% of neurons (160,514) sit in a single strongly connected component. Every central-brain intrinsic, descending, ascending and visual-projection neuron is in it. Only sensory terminals, motor neurons and endocrine cells are outside. 19% of connections and 27% of synaptic weight are reciprocal, meaning both directions exist between the same two cells.

Inhibition is not a correction term

61% of synaptic weight is excitatory (acetylcholine 60%), 38% inhibitory (GABA 21%, glutamate 16%, histamine 0.5%), 0.5% unsigned. The median neuron receives 44% of its input from inhibitory cells, with a 10th to 90th percentile range of 24% to 69%. Central-complex neurons are the most inhibited class at 54%, MBONs (14%) and Kenyon cells (18%) the least.

Reciprocal pairs at neuron level, by sign:

$E \leftrightarrow I$	$E \leftrightarrow E$	$I \leftrightarrow I$
616,156	263,808	126,371

Feedback inhibition is the dominant two-cycle, more than twice as common as recurrent excitation. The bilateral wiring sharpens this. 22% of weight crosses the midline (3% in the optic lobe, 27% in the central brain, 44% in the VNC, 47% for descending and 62% for ascending neurons), and crossing synapses are more often inhibitory (44%) than ipsilateral ones (38%). The strongest left-right mutual inhibition between homologous types is in the central complex: ER4d at 10,156 and 10,083 synapses each way, then Delta7, ER2_c, ER5, ER4m, ER3d_b and ER3m. These are the ring neurons that carry visual and self-motion inputs into the ellipsoid body, and they compete across hemispheres before the heading circuit ever sees them.

Neuromodulatory neurons are broadcasters. The median octopaminergic cell contacts 27 cell types (maximum 1,462), serotonergic 50, dopaminergic 8.

The motif vocabulary, and whether it is surprising

Condense the 165k neurons into 11,752 types and keep type-to-type edges of at least 3 synapses per target neuron and 20 in total. That leaves 473,523 type-level edges. On that graph:

- Feedforward inhibition is the rule, not a motif. Of 277,512 strong excitatory type edges $A \rightarrow B$, 80% have a parallel path $A \rightarrow I \rightarrow B$ through an inhibitory type. The largest central-brain instances are $EPG \rightarrow \text{Delta7} \rightarrow (\text{Delta7}, \text{PEN})$, $\text{Kenyon cell} \rightarrow \text{APL} \rightarrow \text{Kenyon cell}$, $\text{ExR1} \rightarrow \text{ER5} \rightarrow \text{ER5/EL}$, and $\text{ORN_DA1} \rightarrow \text{ILN2F_b} \rightarrow \text{ILN2T_b}$.
- Reciprocal type pairs: 20,816 $E \leftrightarrow I$, 8,272 $E \leftrightarrow E$, 6,375 $I \leftrightarrow I$. The strongest central $E \leftrightarrow E$ pairs are $\text{DPM} \leftrightarrow \text{KC}\gamma\text{-m}$, $\text{KC}\gamma\text{-m} \leftrightarrow \text{PPL103}$, $\text{EPG} \leftrightarrow \text{PEN}$ and $\text{KC} \leftrightarrow \text{PAM}$ dopamine neurons. The strongest $I \leftrightarrow I$ pairs, the winner-take-all candidates, are $\text{ER2_a} \leftrightarrow \text{ER2_c}$, $\text{ILN2F_b} \leftrightarrow \text{ILN2P_c}$, $\text{ER3a_a} \leftrightarrow \text{ER3m}$ and $\text{LNO1} \leftrightarrow \text{LNO2}$. The strongest $E \leftrightarrow I$ is APL with every Kenyon-cell subtype (79,151 and 79,270 synapses with $\text{KC}\gamma\text{-m}$ alone).
- Disinhibition: 78,796 $I \rightarrow I$ type edges. The top central chains are $\text{ER2_c} \rightarrow \text{ER2_a} \rightarrow \text{EPG}$, $\text{ER2_a} \rightarrow \text{ER2_c} \rightarrow \text{EPG}$ and $\text{ER3a_a} \rightarrow \text{ER3m} \rightarrow \text{EPG}$, ring-neuron chains gating the heading bump.
- Feedforward loops, $A \rightarrow B \rightarrow C$ plus a direct $A \rightarrow C$: 415,017 edges in coherent loops (excitatory middle, acting as delay lines and coincidence detectors) and 369,037 in incoherent ones (inhibitory middle, pulse shaping).
- Normalisation cells: inhibitory types whose main input population is also their main output population, so they read a population's total and feed it back. APL (90% in, 94% out Kenyon cells), ILN2F_b and ILN2P in the antennal lobe, Delta7 (99% in, 100% out central complex), the ER ring neurons, and in the VNC the tactile IN01B and gustatory IN05B interneurons.

- Hubs are the same cells. The two APL neurons (119k and 113k input synapses), CT1 (139k output), LPi21, Am1, Li39, DPM. Degree distributions have a heavy tail with exponent about 1.0 in and 1.15 out above 100 synapses. Median in-degree is 260, the 99th percentile 6,301.

Now the important question: are these counts large because the wiring is organised, or just because some types are big? The way to answer is a null model. Rewire the type graph while keeping every type's in-degree, out-degree and the sign of each edge's source, so that only the pairing of pre- to post-synaptic types is shuffled, and count the motifs again. Six rewires give:

metric	observed	null mean $\pm$ sd	z
feedforward-inhibition share of E edges	80%	25% $\pm$ 0.0018	297
reciprocal $E \leftrightarrow E$	8,272	526 $\pm$ 21	370
reciprocal $I \leftrightarrow I$	6,375	282 $\pm$ 15	416
reciprocal $E \leftrightarrow I$	21,602	1,194 $\pm$ 35	587
edges in coherent feedforward loops	415,017	134,573 $\pm$ 705	398
edges in incoherent feedforward loops	369,037	112,842 $\pm$ 1,510	170
$I \rightarrow I$ edges (control)	78,796	78,796 $\pm$ 0	0

Every motif is hundreds of standard deviations above chance. The $I \rightarrow I$ count is conserved exactly by this null, which is why it is there: it is the control that shows the shuffling is doing what it claims. So the answer is yes, the wiring concentrates feedback inhibition, feedforward inhibition and mutual inhibition far beyond what the degree statistics predict. This is the kind of result that a large graph makes easy to get and easy to overinterpret, so I want to flag what the z-scores do and do not say: they say the motifs are not degree artifacts. They do not say what the motifs compute. That is the job of the rest of the book.

The spectral footprint

Treat the signed synapse-count matrix as a linear operator and you get a crude bound on what the dynamics can do. Its spectral radius is 3,771. Of the largest 40 eigenvalues, 32 have positive real part. The dominant eigenvector has a participation ratio of 49.9 cells, and 60% of it sits on one antennal-lobe local-neuron type, ILN1_bc, with the rest on other ILN2 subtypes and a few projection neurons.

The way to read this is: the signed wiring is linearly unstable at many modes, so activity in the real network is shaped by nonlinearity and inhibition, not by passive decay. And the biggest single loop in the brain, as the linear algebra sees it, is local to the antennal lobe. The source-minus-sink axis (each neuron's output weight minus its input weight) separates the broadcasters, which are sensory, modulatory and command cells, from the receivers, which are the motor pools and output neurons that integrate everyone else's vote.

Where structure stops

Everything above was obtained from the graph. A recurring result of the dynamical chapters is that structure systematically overstates what dynamics delivers. Wired inhibitory kernels are realised, but

at a fraction of their structural contrast. Wired phase shifts are realised only with help from recurrence. Wired attractors appear only in narrow gain regimes. Wired normalisation loops sometimes cannot carry their function at count-calibrated weights.

So the right way to think of the graph is as a hypothesis generator of high quality and limited authority. Chapter 4 finishes the structural tour circuit by circuit. Everything after that is about how to test what the graph proposes.

The Circuits, Region by Region

Chapter 3 was about global statistics. This chapter goes through the brain region by region and asks, for each one, what algorithm the wiring looks like it implements. The word “looks like” is doing real work in that sentence. Everything here is a structural signature, and Chapter 5 turns the signatures into a catalogue of operators with confidence scores. But you cannot evaluate the catalogue without seeing the anatomy it came from, so here it is.

Antennal lobe: labelled lines with gain control

The olfactory input stage has 50 glomeruli, 2,562 olfactory receptor neurons (ORNs), 267 uniglomerular projection neurons (PNs) and 420 local neurons (LNs).

Labelled lines. 96.6% of ORN→PN weight stays inside one glomerulus. A glomerulus has a median 35 ORNs and 4 PNs, so a 10:1 convergence. Each PN receives from a median 38 ORNs and each ORN reaches 4 PNs. Odour identity enters the brain as “which glomeruli are active,” and the wiring keeps the channels separate at this stage.

Normalisation. ORNs send more synapses to LNs (537k) than to PNs (414k). The LNs are broad, with a median 8 input glomeruli and 15 output glomeruli, and the top decile spans more than 42 of the 50. They synapse back onto ORN terminals (169k synapses, 41% of the ORN→PN weight), which is the presynaptic gain control that physiology attributes to GABA_B receptors. LN→LN weight (626k) exceeds LN→PN (283k). The LNs are mixed in sign: 115 GABA, 99 glutamate, 137 acetylcholine, 67 unknown.

That last number matters later. The cholinergic LNs, and lLN1_bc in particular, make on the order of 200,000 synapses onto projection neurons. In the whole-brain model (Chapter 14) this is the reason odour channels bleed into each other downstream, and odour specificity in the Kenyon cells is the one calibration target that failed every configuration. The labelled-line picture is what the ORN→PN wiring says. The LN wiring says something messier.

Lateral horn: the innate pathway, at equal weight

Projection neurons split their output between the mushroom body (learnable) and the lateral horn (innate), and the split is even: PN→lateral horn weight is 1.036 times PN→Kenyon cell weight. The horn has 2,028 neurons in 422 types, of which 1,574 are output cells and 454 are local interneurons that repeat the antennal lobe’s normalisation pattern. Each output cell pools a median 6 PN types, with its top PN contributing 25% of its input, so the pooling is broader than a labelled line and narrower than the mushroom body’s near-random sampling. 99% of PNs reach the horn.

The learned and innate channels are not isolated. MBONs synapse back onto the horn, and the horn reaches the descending bottleneck directly, most heavily onto DNp32 (2,322 synapses), pIP1 (2,072) and DNp06 (1,214).

Mushroom body: random expansion, global feedback, gated readout

This is the circuit that reads like a machine-learning diagram, which is why Chapter 11 tests it so carefully.

- Expansion. 4,064 Kenyon cells (KCs) from 686 PNs, a $5.9\times$ expansion ($15\times$ against the 267 uniglomerular PNs). Each KC samples a median 5 PNs (10th to 90th percentile 3 to 8) through its dendritic claws. 282 PNs reach the calyx, each contacting a median 56 KCs. 283 KCs have no PN input at all: 185 γ -d and 87 $\alpha\beta$ -p cells, the subtypes that take visual rather than olfactory input.
- Near-random sampling. The mean absolute correlation between PN types, over which KCs they contact, is 0.023. Shuffling the KC targets gives 0.013. So there is a small structured excess and the rest is random, which is what a random-projection memory wants.
- Sparsening. Every KC drives the giant interneuron APL and every KC is inhibited by it: KC $\rightarrow$ APL 210k synapses, APL $\rightarrow$ KC 196k, 100% coverage both ways. KC $\rightarrow$ KC axonal synapses (491k) actually exceed PN $\rightarrow$ KC (389k).
- Readout. 97 MBONs, each reading a median 313 KCs (maximum 1,680). Each KC reaches a median 11 MBONs. KC $\rightarrow$ MBON is 439k synapses.
- Three-factor teaching. About 340 dopaminergic neurons (DANs) synapse onto KC axons (89k) more than onto MBONs (38k), and 70% of DAN $\rightarrow$ MBON weight lands in the matching compartment. MBON $\rightarrow$ DAN (9k) and MBON $\rightarrow$ MBON (25k) close a loop that lets one compartment's output set another's teaching signal.

Where the compartments' votes recombine. The compartmental readout is not the end. MBONs of different compartments and opposite valence converge on shared targets: 20 targets are fed by two or more MBON types, and a direct channel of 4,073 synapses reaches the descending neurons. Same-sign MBONs share more of their target repertoire (cosine 0.026) than opposite-sign pairs (0.02), so the valence channels stay separable where they meet. The MBONs themselves are 26 glutamate, 50 acetylcholine and 21 GABA.

Central complex: a ring with a shifter, and vector arithmetic behind it

Protocerebral-bridge glomeruli in the instance names give each EPG, PEN, PEG and Delta7 neuron a column, and the heading ring has period 8 per side.

The cosine inhibition kernel. Two-hop EPG $\rightarrow$ Delta7 $\rightarrow$ EPG weight against column offset (mod 8):

offset	0	1	2	3	4	5	6	7
synapses	115	184	830	1,510	1,770	1,560	828	190

Minimum at the bump, maximum opposite it. That is the $1 - \cos \Delta\theta$ inhibition of a ring attractor, and it fits $a - b \cdot \cos$ with $a = 873$, $b = 890$, $R^2 = 0.99$ and contrast 0.878. Each Delta7 reads a median 32 EPGs and writes onto 8, and Delta7 cells are 99% central-complex in and out. The profile holds across hemispheres too, with left glomerulus L_k paired with R_(9-k) and R_(10-k).

Local recurrent excitation. EPGs of one column synapse onto each other (33 synapses per pair at offset 0, 3 at offset ± 1) and reciprocally with PEG (93 per pair at offset 0).

The shifter. EPG $\rightarrow$ PEN goes to column +1 and PEN $\rightarrow$ EPG returns to -1 and -2 on the same side. Split by hemisphere the loop is asymmetric, and resolving each PEN's EPG targets on the ring gives a clean bimodal answer:

population	p10	median	p90
PEN (both types), left	+1.42	+1.47	+1.55
PEN, right	-1.55	-1.45	-1.37

Left-hemisphere PENs write about 1.5 columns ahead, right-hemisphere PENs about 1.5 behind, and PEN1 and PEN2 agree. The pooled median is near zero only because the two hemispheres cancel. So the two PEN populations push the bump in opposite directions, which is angular-velocity integration. One caveat: the analysis is at glomerulus resolution (45°), and the real shift is one wedge (22.5°), so the shift appears as an asymmetry in the ± 1 bins rather than as a clean peak.

Ring-neuron competition. The ER types that feed the ring show the strongest mutual inhibition in the brain, across hemispheres and between subtypes (ER2_a $\leftrightarrow$ ER2_c at 5.8k and 6.1k synapses), plus disinhibitory chains onto EPG. Inputs compete before they reach the attractor.

Fan-shaped body vector shifts. Between column-tagged FB cells, 201k synapses are in type pairs whose peak offset is the same column, and 146k in pairs peaking 3 or more columns away. hDelta outputs peak 4 to 5 columns away, which is half of 8 to 9 columns, a 180° shift, onto PFL3, PFR, vDelta, FC2B, FC2C and PFGs. PFNd $\rightarrow$ hDelta peaks at -3 . This is the circuit Chapter 10 tests. The steering readout, PFL3 and PFL2, projects onto the descending neurons DNa03 (2,039 synapses), DNb01 (753) and DNa02 (736).

Optic lobe: convolution, ON/OFF, a filter bank, and pooling

- Shared kernels. 47 cell types have 600 or more copies, one per column per eye. For the strongest columnar type pairs, 99 to 100% of source neurons are connected, with a fixed number of partners and a coefficient of variation of total weight of 0.2 to 0.36. That is one kernel applied at every column. One-to-one kernels include L1 $\rightarrow$ Mi1, L1 $\rightarrow$ L5, L2 $\rightarrow$ Tm1, L2 $\rightarrow$ Tm2. Five-to-six-column kernels include L1 $\rightarrow$ Tm3, Mi1 $\rightarrow$ Tm3 and Mi1 $\rightarrow$ T4a to d.
- ON/OFF split. L1 sends 142k synapses to Mi1 and 147k to Tm3 and nothing to the OFF cells. L2 sends 205k to Tm1, 220k to Tm2, 160k to Tm4, 158k to T1 and nothing to Mi1 or Tm3. The lamina output matrices are block-diagonal.
- A direction filter bank. The four T4 subtypes have the same input composition (Mi1 about 32%, Tm3 13%, Mi9 10 to 12%, CT1 6 to 7%, Mi4 5 to 6%), with cosine similarity 0.93 to 0.95 between their full input vectors. They are four copies of one kernel differing only in spatial offset. T5a to d likewise. CT1, the top output hub in the brain, gives feedforward inhibition to every T5 subtype.
- Pooling. Wide-field lobula-plate cells sum one subtype each. HSE, HSN and HSS each receive from about 400 to 540 T4a and T5a cells and no other subtype. H2 pools 760 T4b/T5b. Thousands of local detectors feed one integrator.

The visual feature menu. 9,203 visual projection neurons in 346 types carry the optic lobe's features to the central brain. Each channel is focused (median 4 target types at 5% or more) and the channels

are nearly non-redundant (median pairwise target-profile cosine 0.002, 90th percentile 0.05), but downstream they recombine: 275,710 synapses onto descending neurons, 23,508 onto the lateral horn, 1,250 onto the compass ring neurons.

Escape: a three-synapse funnel

The giant fibre receives 36k input synapses. LC4 (126 neurons, 55 to 71 converging on each giant fibre) provides 17.6% and LPLC2 (182 neurons, about 90 per giant fibre) 13.4%. The rest is spread over hundreds of types. Photoreceptor to giant fibre is 3 hops at 5 or more synapses. Its outputs go to GFC2 to 4, the jump motor neuron TTMn, DNp11 and PSI.

The descending funnel

1,314 descending neurons (DNs) receive only 3.7% of all brain output synapses. The median DN gets 1,722 synapses from 87 types (10th to 90th percentile 335 to 7,883 synapses, 26 to 253 types). DNs talk to each other: 31,993 connections, 6,801 of them reciprocal. Only 6% of DN output goes directly to motor neurons. Half goes to VNC interneurons. And the return path is as heavy as the command: ascending neurons send 527k synapses back onto DNs against 381k DN→AN. The largest DNs by input are DNp103, DNa02, DNp06, DNp16, the giant fibre DNp01, pIP1 and DNp100.

Motor pools

708 motor neurons. The median one has 143 premotor neurons and 2,464 premotor synapses, 44% of them inhibitory, 88% from VNC interneurons, 9% from descending neurons, 2% from sensory. Motor neurons of one type and side share premotor partners with a median Jaccard of 0.03 against 0.002 for random pairs, a 15× excess: these are pools. 43% of VNC weight crosses the midline and 53% of that is inhibitory. The strongest reciprocal inhibitory type pairs (IN16B049↔INXXX217, IN13A001↔IN19A001, IN08A002↔IN19A011) are half-centre candidates.

State cells are threaded through, not layered on top

A small set of identified cells carries brain state: 40 circadian clock neurons (DN1a, DN1p, s-LNV, LNd, LPN), 49 sleep-need cells (ER5, ExR1, ExR2, hDeltaC), 37 octopamine and 8 serotonin broadcasters, and 53 peptidergic cells. They are few, but their wiring is dense where it matters:

link	synapses
central complex → state	111,308
state → central complex	101,159
state → state	37,036
state → descending neurons	8,686
state → MBON	3,392

The two largest flows are a two-way loop with the compass. State is not a layer above the fast circuits. It is wired into them, which is one reason Chapter 14's neuromodulation results come out the way they do.

The summary table

structure	where	evidence
labelled lines plus divisive normalisation	antennal lobe	97% within-glomerulus, broad LNs, LN→ORN presynaptic inhibition
random expansion plus global feedback sparsening	mushroom body	$5.9\times$ expansion, 5 claws, near-random sampling, APL 100% coverage
compartmental readout with three-factor plasticity	mushroom body	97 MBONs $\times$ 313 KCs, DAN→KC $\gg$ DAN→MBON, 70% compartment match
ring attractor with cosine inhibition	ellipsoid body and bridge	Delta7 profile 115 $\rightarrow$ 1,770 $\rightarrow$ 190 over half a turn
angular-velocity integrator	PEN	± 1.5 -column push through left and right PENs
vector arithmetic by half-ring shift	fan-shaped body	hDelta outputs 4 to 5 columns away
shared convolution kernels	optic lobe	47 columnar types, fixed partner counts, CV 0.2 to 0.36
parallel ON/OFF channels	lamina to medulla	block-diagonal L1/L2 output
direction filter bank plus wide-field pooling	T4/T5 to lobula plate	four identical kernels, HS pools only subtype a
feedforward inhibition everywhere	whole CNS	80% of strong E edges have a parallel I path
winner-take-all inputs	ring neurons, antennal-lobe LNs	strongest I $\leftrightarrow$ I pairs, cross-hemisphere mutual inhibition
sensorimotor bottleneck with loops	descending neurons	1,314 DNs, 3.7% of brain output, AN→DN $\approx$ DN→AN
motor pools with commissural inhibition	VNC	$15\times$ shared premotor input, 53% inhibitory crossing

Does the structure survive dynamics? A first check

Chapters 7 to 11 do this properly, but there is a cheap version to show now. Take the calibrated whole-brain LIF model of Chapter 14, with the same graph and sign conventions, and probe three of the structures above:

probe	structural claim	measured in the LIF model
Delta7 kernel	cosine, R^2 0.99	fire each Delta7 and read the inhibitory conductance on the ring: minimum at the bump wedge, maximum opposite, cosine R^2 0.56

probe	structural claim	measured in the LIF model
PEN shifter	± 1.5 columns	drive one column's PENs: EPG wedges depolarise +1.52 columns for left PENs, -1.48 for right
APL sparsening	global feedback	same PN drive, APL silenced: active KC fraction $0.15 \rightarrow 0.72$, mean rate $0.87 \rightarrow 5.08$ Hz

All three pass in sign. But notice the Delta7 number: the kernel that fits the wiring at R^2 0.99 fits the realised inhibition at R^2 0.56. Same graph, same cells. The shape survives, the precision does not, and the calibrated LIF does not sustain a freely rotating bump at all without external drive. That gap between 0.99 and 0.56 is a preview of the book's main theme.

The Operator Layer: An Instruction Set for a Brain

From signatures to a catalogue

Chapters 3 and 4 produced a long list of structural facts. The operator layer compresses that list into something you can actually reason about: a catalogue of candidate computational operators, which the project calls a BioISA, by analogy with an instruction set architecture. The idea is that a nervous system, like a processor, has a small vocabulary of primitives that it composes, and the vocabulary is what you want to extract.

Each catalogue entry has the same shape:

- a substrate: which cell types, and how many cells.
- a signature: the wiring pattern that triggered the detection.
- semantics: what it takes in, what state it holds, what it puts out.
- a conventional equivalent: the nearest thing in an engineering textbook.
- a reduced model: a few lines of pseudocode.
- evidence items with weights, counterevidence, and predictions.
- a confidence score.

The scanner reads the structural report and emits twelve entries. Nine are distinct:

operator	region	confidence	substrate and signature
ring_attractor	central complex	0.70	EPG, Delta7, PEG, PEN (88 cells); columnar ring, two-hop inhibitory kernel $\propto 1 - \cos \theta$, mirror-symmetric shifter
winner_take_all	motif, many regions	0.60	6,375 I $\leftrightarrow$ I reciprocal type pairs, $z = 416$ against the null

operator	region	confidence	substrate and signature
phasor_vector_shift	central complex	0.55	PFN $\rightarrow$ h Δ B with $-3 / +2$ column offsets; h Δ B fans out at $+4/+5$ columns to 13 target types
motion_correlator	optic lobe	0.55	T4/T5 subtypes with identical input composition (cosine 0.94) and spatial offsets; ON/OFF split
convolution_front_end	optic lobe	0.55	47 columnar types, fixed partner counts, weight CV 0.19 to 0.47
sparse_associative_memory	mushroom body	0.55	$5.9\times$ expansion, near-random sampling, APL feedback covering 100%, DAN $\rightarrow$ KC $\gg$ DAN $\rightarrow$ MBON
feedforward_inhibition	motif	0.55	80% of strong E edges have a parallel I path, $z = 297$
divisive_normalization	antennal lobe, mushroom body	0.50	30 candidate cells whose main input population equals their main output population
command_funnel	descending	0.30	1,314 DNs carrying 3.7% of brain output; AN $\rightarrow$ DN loop as heavy as DN output

The remaining three entries are low-confidence re-detections of the same motifs at other thresholds (0.25, 0.25, and a 0.0 ring-attractor candidate that fails its own checks). They are kept because the catalogue should show what the detector *almost* fired on, not just what it accepted.

What a detection means, and what it does not

I want to be careful here, because “the connectome contains a ring attractor” is the kind of sentence that gets repeated without its qualifier.

A detection is a *signature match*. It is not a claim that the circuit computes anything. `ring_attractor` scores 0.70 because its signature is unusually specific: a columnar population whose two-hop inhibitory kernel fits a cosine at R^2 0.99 (weight 0.35), with contrast 0.878 (weight 0.15), a mirror-symmetric shifter at $L + 1.47$ and $R - 1.45$ columns (weight 0.2), plus two dynamical checks from the whole-brain model that the kernel and the push field are realised (0.15 each). There are not many things a circuit with that

wiring could be other than a continuous attractor with velocity input.

`command_funnel` scores 0.30 because “a lot of fan-in onto a small population” is consistent with many computations, and the catalogue says so in its counterevidence field: the funnel count mixes cell types with very different functions, so it may be a bottleneck of bandwidth rather than of decisions.

The scores are hand-weighted sums of evidence items, and the weights are judgment calls. I would not defend 0.55 versus 0.60 as meaningful. I would defend the ordering, and I would defend the requirement that every entry lists what would prove it wrong.

Every entry has to say what would falsify it

This is the part of the catalogue I think is most valuable, so here are the predictions as written:

- `feedforward_inhibition`: blocking the inhibitory arm should broaden the temporal tuning of the target. Reduced model: $y = x[t] - g \cdot x[t-1]$.
- `winner_take_all`: strong simultaneous drive to both sides should produce bistability or oscillation rather than a compromise. Reduced model: $a \leftarrow g \cdot b$; $b \leftarrow g \cdot a$.
- `convolution_front_end`: a feature learned at one eccentricity should transfer to every eccentricity. Counterevidence already recorded: some pairs have CV near 0.36, so it is not a perfect convolution.
- `sparse_associative_memory`: Kenyon cells should be sparse and decorrelated, and APL should implement gain control. Conventional equivalent: locality-sensitive hashing plus a linear readout, trained by a three-factor Hebbian rule rather than backpropagation. Chapter 11 is what happened when the gain-control half of that prediction was tested.
- `command_funnel`: silencing a single DN type should remove a specific motor program. Chapter 14 does this for several.
- `ring_attractor`: silencing Delta7 should change bump width, and unilateral PEN silencing should abolish integration in one direction only. Chapters 7 to 9.

Signature versus semantics

The catalogue keeps two things apart on purpose:

- The signature is what the wiring looks like: a cosine kernel, a column offset, an expansion ratio. It is structural, measurable, and reproducible from the graph.
- The semantics is what the circuit computes: a heading phase, a rotated vector, an odour identity. It is a hypothesis, to be established dynamically.

There is also a `topology_folded` flag for cases where the graph-theoretic form hides a geometric one. The ring attractor is *not* folded: the ring is a real ring in the wiring. The convolution front end and the phasor shift *are* folded: the graph alone does not show a lattice or a column axis, and you need the positional labels from the instance names to see them.

From an operator to an experiment

Each catalogue entry is the input to the executable stage, and the mapping is mechanical enough to describe in one paragraph. The substrate names the populations to instantiate. The free quantities, the gains that the wiring does not fix, become the axes of an ensemble. The predictions and counterevidence

become the perturbations. Chapters 7 to 11 follow three operators through that process by hand, and Chapter 13 describes the machinery that makes a fourth operator a matter of writing a spec entry rather than a new script.

The Ensemble Method: Why the Wiring Gives You a Family, Not a Model

The underdetermination is specific and enumerable

Here is the basic problem. A connectome tells you the graph: which cells connect, with how many synapses, and what sign the presynaptic transmitter implies. It does not tell you:

1. the conductance each synapse produces, meaning the count-to-PSP calibration.
2. the strength or even the existence of gap junctions.
3. the neuromodulatory state, which changes thresholds over seconds.
4. thresholds, adaptation, and resting-potential heterogeneity across cells.
5. synaptic delays and time constants beyond a single global value.

The naive approach is to pick values for all of these, tune them until the model does something reasonable, and publish the model. The problem with the naive approach is that the tuning is where the answer gets smuggled in. If you tune the heading circuit until it holds a bump, you have not learned that the wiring supports a bump. You have learned that you can make it hold one.

The alternative, which is what the project does, is to construct the set of models consistent with the wiring: an ensemble that spans the free parameters on a grid, with every member sharing the exact same graph. Then ask which conclusions hold across the whole family, and where the family splits. This inverts the usual logic. Instead of fitting one model to data, structure defines the family and you measure agreement.

The disagreement is the product. If two members agree at baseline but disagree about what happens when population X is silenced, then “silence X” is a *discriminating experiment*, and you can rank it against every other candidate experiment by how much of the family it splits.

The neuron model, and where its numbers come from

Every ensemble member is a network of leaky integrate-and-fire (LIF) neurons over exactly the circuit’s cells. The model follows Shiu et al. 2024: rest -52 mV, threshold -45 mV, membrane time constant 20 ms, synaptic time constant 5 ms, 0.5 ms steps, 1.8 ms synaptic delay. Synapses are conductance-based, with excitatory reversal at 0 mV and inhibitory reversal fitted near -76 mV, because current-based synapses let any neuron with 150 synapses fire any target and that is not how membranes work.

The global parameters that the whole-brain model of Chapter 14 fitted against published behaviour are inherited here:

parameter	searched range	fitted
synaptic strength per synapse	0.2 to 1.2 mV	0.55 mV
neuron-size exponent ($\text{PSP} \propto (\text{volume} / \text{regional median})^{-\alpha}$)	0 to 1	0.61
Kenyon-cell threshold offset	0 to 30 mV	10.9 mV
inhibitory gain	0.5 to 5	0.61
inhibitory reversal	−85 to −55 mV	−76.4 mV
minimum synapses per connection	3 to 10	6
adaptation	0 to 3 mV	0
refractory period	2 to 6 ms	3.8 ms

Sign comes from the presynaptic transmitter, and unknown contributes nothing. The ensemble then scales named *edge classes* (EPG→EPG, Delta7→EPG, and so on) by its free parameters on top of this base. So a member with `epgRecur = 4` has EPG recurrence at four times the count-calibrated strength. The base calibration fixes the units. The ensemble explores the multipliers the wiring leaves open.

The machinery

Three generic pieces are shared by every circuit lab:

- A builder. Given the circuit’s cells, its edges, and a set of named parameter axes, the builder returns a factory from parameter values to a LIF network. Edge classes are specified as `{pre: type or set, post: type or set, param: name}`, and tonic bias populations as `{pop, param}`.
- A seeded random number generator (mulberry32). Every probe, whether an odour pattern, a noise trace, or an initial bump, is reproducible from the seed. Ensemble comparisons are only meaningful at fixed randomness, and conclusions are only trusted when the *ranking* of experiments survives reseeding, not just the scores.
- Perturbation helpers: silencing a population by clamping its threshold, zeroing an edge class, scaling a drive.

The protocol

Every lab follows the same six steps.

1. Grid. Sweep the free axes. The ring lab uses $\text{epgRecur} \in \{1,3,4,6\} \times \text{d7Gain} \in \{0.5,1,1.3\} \times \text{epgTonic} \in \{0,3,5,7\} \times \text{penGain} = 1$, which is 48 members. The memory lab uses $\text{kc2mb} \in \{0.5,1,2\} \times \text{aplGain} \in \{0.5,1,2\} \times \text{mbonTonic} \in \{0,4\} \times \text{mbRecur} = 1$, 18 members.
2. Probe. Drive every member with the same stimulus and measure observables that fit the circuit’s geometry. On a ring that is bump phase, width and persistence. On a column map it is centroid offset. On a memory it is pattern separation and gain compression.
3. Classify. Label each member with a hypothesis class. For the ring: `silent`, `attractor_tonic`, `attractor_free`, `filter`, `frozen`. For the shift: `wired_shift`, `passthrough`, `silent`. For the memory: `gain_controlled`, `linear_passthrough`, `collapsed`, `silent`.

4. Perturb. Repeat the probe under each perturbation: silence a population, remove an edge class, sweep a drive.
5. Split. For each experiment, compute the *separation score*: the fraction of member pairs the experiment places in different outcome classes.
6. Rank. Order experiments by separation. The top one is where the surviving hypotheses disagree most, so it is the measurement most worth making.

The output is a JSON artifact with the seed, the grid, every member's parameters and outcomes, the class counts, the mechanism counts, the ranked experiment table, and a one-paragraph claim. The claim for the ring lab reads:

Wiring supports a low-dimensional heading bump in a narrow gain regime (EPG recurrence ~ 3 to $6\times$, tonic EPG bias ~ 3 to 7 mV). Free-running bump without tonic drive was not observed at any grid point. Among bump-sustaining models, Delta7 silencing separates mechanisms that are indistinguishable at baseline.

Notice the grammar. It is a claim about the *ensemble*, not about a model. It reports a regime, a non-observation, and the experiment that discriminates. That sentence shape is the deliverable of the whole pipeline.

What a separation score is

The score is simple to compute and better understood than treated as a black box. For an experiment that sorts 48 members into outcome classes, count the pairs of members that land in different classes and divide by the total number of pairs. If every member responds the same way the score is 0. If the members split evenly between two classes it is 0.5. With three or more balanced classes it can go higher.

The consequence is that a score near 0.5 on a small ensemble is close to the maximum, and a score near 0.53 in the mushroom-body lab with only two occupied classes (Chapter 11) is a near-even split, not evidence of a rich mechanism landscape. The scores are only comparable within one ensemble. *tonic_sweep* scored 0.414 in the ring lab because members in different mechanism classes answer a tonic-drive sweep with categorically different bump behaviour. That is what “best discriminator” means: not the biggest effect, but the most informative outcome about which member you are looking at.

What the method cannot do

An ensemble is only as good as the mechanisms it contains. If the family of models lacks a mechanism the real circuit has, no amount of grid points will find it. The family will split into classes that are all wrong in the same way, and the ranked experiment will discriminate between wrong answers. Chapter 9 is a worked example of this, and of how the missing mechanism was found: not by more members, but by a second formalism and by published data.

The Heading Circuit as a Spiking Ensemble

The first operator to go end-to-end is `ring_attractor`. It is the best-understood computation in any connectome, which is why it goes first: if the method cannot recover a known answer, its answers elsewhere are worthless.

The circuit

88 cells in five populations:

- EPG (46 cells): the ring population. Bump position encodes heading.
- Delta7 (42): the inhibitory surround. Each reads a median 32 EPGs and writes onto 8, and the two-hop $\text{EPG} \rightarrow \text{Delta7} \rightarrow \text{EPG}$ kernel is the cosine from Chapter 4.
- PEG (18): a copy population, reciprocally wired with EPG in the same column.
- PEN_a and PEN_b (20 and 22): the shifter. $\text{EPG} \rightarrow \text{PEN}$ goes one column ahead, $\text{PEN} \rightarrow \text{EPG}$ returns one to two columns behind, and the net push resolves to about +1.5 columns for left-hemisphere PENs and -1.5 for right. They carry the angular-velocity signal.

Structurally this is the Amari ring recipe almost verbatim: local excitation, a cosine surround, and a mirror-symmetric shifter. The kernel fits a $-b \cdot \cos$ at R^2 0.99 with contrast 0.878. The question is whether the dynamics deliver what the structure advertises.

The ensemble

Four free axes, 48 members:

axis	what it scales	why it is free	grid
epgRecur	$\text{EPG} \rightarrow \text{EPG}$	recurrence gain is not fixed by synapse counts	1, 3, 4, 6
d7Gain	$\text{Delta7} \rightarrow \text{EPG}$	Delta7 inhibition is glutamatergic via $\text{GluCl}\alpha$; effective strength uncertain	0.5, 1.0, 1.3
epgTonic	tonic EPG bias (mV)	resting excitability is unknown	0, 3, 5, 7
penGain	$\text{PEN} \rightarrow \text{EPG}$	shifted-feedback gain	1

Each member is given a seeded bump, released, and measured for persistence (concentration and full

width at half maximum), for the realised Delta7 kernel, and for bump rotation under unilateral PEN drive. Perturbations: silence Delta7, lesion PEG, sweep EPG tonic excitability, drive one PEN arm.

Finding one: the attractor exists, but only in a corner of the grid

class	members	meaning
silent	30	no sustained bump at any drive
attractor_tonic	12	bump persists, but only with tonic EPG bias of 3 to 7 mV
filter	6	activity follows the input, no self-sustained bump
attractor_free	0	bump persists with no tonic drive

That last row is the one to look at. A free-running bump, the thing the textbook picture of a ring attractor promises, was not observed at *any* grid point. Every bump-sustaining member needs a depolarising bias to cross into the bistable regime, and the viable region is recurrence 3 to 6 $\times$ count-calibrated strength with 3 to 7 mV of tonic bias, roughly 15 to 25% of the grid.

So the wiring supports the attractor's geometry, but the dynamics need an excitability floor the wiring does not provide. The fly's heading system, on this evidence, is something that must be switched on rather than something that is always on. That is consistent with the whole-brain model, where the calibrated LIF also fails to sustain a free bump without external drive (Chapter 4).

Finding two: Delta7's role splits the survivors

Among the 12 attractors, silencing Delta7 gives three different outcomes:

mechanism class	members	what silencing Delta7 does
d7_sculpts_sharp	8	bump survives and stays sharp; Delta7 sharpens but is not required
d7_essential	3	activity extinguishes
d7_confines	1	bump survives but widens

All twelve look the same at baseline. You cannot tell them apart by imaging a fly that is walking normally. You can tell them apart by one perturbation. This is the discriminating-experiment logic doing what it was built for.

One caveat that the artifact carries and I want to repeat: the d7_essential outcome, where released excitation drives an adaptation shutdown, is flagged as a possible LIF artifact rather than a biological prediction. The literature also argues against it. Real Delta7 $\rightarrow$ EPG inhibition is glutamate acting on GluCl α and suppresses EPGs *far from* the bump, which is the antipodal kernel we measured, so the sculpting and confining outcomes are the plausible ones. That makes d7_essential the class real data should rule out, which is what a discriminating experiment is for.

The ranked experiments

Separation scores at seed 42:

experiment	score	what it discriminates
tonic_sweep	0.414	attractor versus filter versus silent: the regime boundary
peg_lesion	0.370	whether the PEG copy pathway is load-bearing
d7_silence	0.351	the three Delta7 mechanism classes
unilateral_pen_R	0.284	PEN arm gating, velocity direction
unilateral_pen_L	0.120	the two PEN arms are not equivalent

The top three are stable across seeds, though their order shuffles. Each maps to a real experiment: depolarise EPGs while imaging the ring, kill the PEG copy, silence Delta7 and watch bump width. The asymmetry between the two PEN arms in the last two rows is real in the wiring (the left-through-PEN loop weight ratio is 1.7:1 and the right 1.3:1) and the ensemble picks it up.

The generic runner reproduces this

When the same circuit was later run through the declarative spec machinery of Chapter 13 instead of the hand-written lab, seed 42 gave 30 silent, 11 attractor-tonic and 7 filter members against the lab's 30, 12 and 6, with the same best experiment (`tonic_sweep`, separation 0.393 versus 0.414) and the same top-three set. One member moved across a class boundary. I mention the discrepancy because it is the size of discrepancy you should expect when two implementations share a graph but not a code path, and because “the compiler reproduces the lab” should mean this and not something vaguer.

What the ensemble could not say

Here is the honest limit. The LIF members classified Delta7 as either essential or sculpting. Eight of twelve said the bump survives silencing *sharp*. Only one said it survives *widened*, and Chapter 9 will show that the widened outcome is the one the published data supports.

The ensemble did not fail, exactly. It reported a split rather than a false consensus, and its minority class was right. But its majority was wrong, and the reason is the limit stated in Chapter 6: the family only contained one inhibitory channel. The missing mechanism, a second inhibitory source that is not Delta7, was found by the field model and by the literature, not by more grid points.

The Continuous Field: Deriving the Mechanism

The LIF ensemble answers “does the wiring support an attractor.” A different question, “what mechanism moves the bump,” wants a different formalism. Instead of 46 discrete EPG cells, model the ring as a continuous activity field on the circle, in which the bump is a solution of a partial differential equation rather than a property of individual neurons. This is an Amari neural field, and it is the second of the two formalisms the project uses on the same circuit.

The model

A ring of n units (256 by default, over 8 columns per side) carries an activity field $u(\theta)$:

$$\tau \frac{\partial u}{\partial t} = -u + W_{d7} * f(u) + \text{recur} \cdot K_{\text{exc}} * f(u) + \text{pen_input} + \text{landmark_input} - \text{rn} \cdot \text{mean}(f(u))$$

The point of the model is that every term is measured or derived, not invented:

- W_{d7} is the Delta7 kernel, the fitted $a - b \cdot \cos$ with $a = 873$ and $b = 890$.
- K_{exc} is a narrow local-excitation kernel standing for the EPG/PEG recurrence. Its gain, recur , is the one free parameter.
- pen_input is the velocity pathway, in several formulations discussed below.
- landmark_input is ring-neuron-style disinhibition: global suppression with a gap at the landmark bearing, which anchors the bump to a cue.
- $\text{rn} \cdot \text{mean}(f(u))$ is a second, shallow inhibitory channel that scales with total bump mass. It was not in the first version of the model. Chapter 9 explains why it was added.

f is a saturating nonlinearity. Without it, recurrent excitation runs away, and real neurons have rate saturation, so the model has to as well.

Deriving the velocity gain instead of tuning it

The PEN pathway shifts the bump during rotation. The wiring says “shift by about 1.5 columns per side.” It does not say at what *gain* the shift should be applied, and earlier versions of the project tuned that number by hand. The field model lets you derive it.

Write the bump as a travelling profile $u * (\theta - \varphi(t))$ and the PEN arm as an extra shifted kernel, $v \cdot K_s * f(u)$. Because the field is translation-invariant, the bump has a marginal direction: you

can slide it around the ring at no cost. Project the PEN perturbation onto that direction, which is the derivative of the bump profile u^* , and you get the drift rate in closed form:

$$-\tau \dot{\phi} \langle u^*, u^* \rangle = v \langle u^*, K_s * f(u^*) \rangle$$

$$\rightarrow \dot{\phi} = v \cdot \langle u^*, K_s * f(u^*) \rangle / (\tau \langle u^*, u^* \rangle)$$

Both inner products are numbers you can compute from the kernels, so the integration gain is a derived property of the wiring: $\text{pen_gain} \approx 0.27$, and the theory predicts a drift of about 3.7 per unit drive. There is one subtlety that mattered in practice. The inner products have to be evaluated on an *equilibrated* bump. Calibrating on the unrelaxed seed profile gave a wrong gain and a frozen benchmark, and it took a while to find that.

With the literal shifted-synapse kernels at the predicted gain, the field tracks clean velocity at about $0.85\times$ the ideal. The 15% deficit is a second-order effect: the shifted input also distorts the bump's shape, which the leading-order projection ignores.

Four ways to move a bump, and which ones survive noise

The real test is fluctuating velocity, because a mechanism that integrates clean input and falls apart under noisy, sign-flipping input is not the fly's mechanism. Four formulations of the PEN pathway were implemented and run through the noisy benchmark on the single-inhibition-channel model:

mechanism	how velocity enters	noisy RMS error (rad)	what happened
shifted	amplitude coding: ω scales a shifted injection into EPG	1.29	each injection distorts the bump shape, not just its phase
shifted2	amplitude coding through a second attractor field	1.61	a second field does not repair it
shifted3	position coding: ω displaces a PEN bump, EPG is pulled toward it	0.60	works, and predicts an EPG-PEN phase offset
advect	idealised transport of u itself	0.41	the bound the mechanisms approach

Amplitude coding fails under sign-flipping drive no matter how it is filtered (the PEN time constant was scanned from 0.02 to 0.3 s) and no matter whether a second field is added. Position coding halves the error and is robust, but it plateaus at about 0.60 because the EPG bump is always chasing the PEN bump with one stage of lag. Direct advection is the limit when transport acts on the transported field itself.

Two things fall out of this that are more interesting than the ranking.

First, the 50% gap between shifted3 and advect is the price of indirect transport. The fly, if it uses position coding, pays for a real mechanism with lag.

Second, and more important, position coding makes a *measurable* prediction that amplitude coding does not: during rotation the EPG and PEN bumps should be offset in phase. That is what calcium

imaging of the real circuit shows. So the model discrimination is not just “one of these has lower error.” It is “one of these has a neural signature you can look for, and people have looked, and it is there.”

Why two formalisms

The LIF ensemble and the field model are not redundant, and it would be a mistake to keep only the one that is easier to run.

The field model can express a graded curve: bump width as a continuous function of Delta7 suppression. The LIF ensemble can only vote among classes. The LIF ensemble respects the actual cell-level wiring, including the PEN arm asymmetry and the fact that there are 46 EPGs and not a continuum. The field model smooths that over but exposes the mechanism’s continuous limit and lets you derive gains instead of tuning them.

When they agree, you have agreement across model classes, which is much stronger than agreement across parameters within one class. When they disagree, you have the subject of the next chapter.

When the Two Models Disagree, and Biology Decides

The strongest test the project has produced did not come from a new model. It came from a disagreement between two existing ones, resolved against data that the pipeline had not seen. I think this chapter is the clearest demonstration in the book of why the method is built the way it is.

Running the LIF lab's experiments on the field model

The perturbations that the spiking ensemble ranked highest were run on the field model, so that outcomes could be compared across formalisms:

- `d7_sweep`: graded Delta7 suppression, measuring bump width and amplitude R .
- `pen_both`, `pen_left`, `pen_right`: PEN arm gating.
- `exc_sweep`: the one free excitation gain, to find the bistability threshold.

Two of these validated the machinery cleanly. With only the left PEN arm intact the field drifts 0.98 rad in 2 s, and with only the right arm it drifts exactly 0. Arm selectivity is exact: unilateral PEN loss abolishes integration in one direction only, which matches the fly experiment where PEN block stops the heading from following turns. And the excitation sweep found the bump absent at `epg_recur` 2.6 and below and present at 3.0, so the free parameter's operating point is now bracketed rather than assumed.

The Delta7 disagreement

The graded sweep produced a smooth curve:

Delta7 gain	width (FWHM)	R
1.0	90.5°	0.597
0.8	94.1°	0.601
0.6	98.1°	0.605
0.4	104.6°	0.605
0.3	110.1°	0.599
0.2	116.1°	0.592
0.1	126.8°	0.571
0.0	143.1°	0.528

The bump *survives* complete Delta7 block, widened from 90° to 143° and weakened, but formed. In the LIF lab’s vocabulary that is `d7_confines`, the minority class with one member out of twelve. The majority class, eight of twelve, said the bump survives *sharp*. So the field model and the spiking ensemble disagree, and the artifact records it: `agrees_with_lif_majority: false`.

Checking against data the pipeline had not seen

Turner-Evans et al. (2020) had already done the real experiment. Their finding:

The E-PG population organizes into a single bump even if the output of the $\Delta 7$ neurons is reduced... the bump no longer reliably tracks the fly’s movements.

The real bump survives, and tracks badly. That rules out the LIF majority. But it also rules out the *first version* of the field model, which I have not mentioned yet. That version attributed all surround inhibition to Delta7, and below a Delta7 gain of about 0.3 it dissolved the bump entirely. The curve above is from the corrected model.

The correction was a mechanism, not a parameter

It would have been easy to fix the first field model by turning a knob until the bump survived. That is not what was done, and the difference matters.

Turner-Evans et al. say, in the same paper, that “other sources of inhibition must act.” The connectome agrees: Chapter 4 lists the GABAergic ring neurons as the strongest mutual-inhibition population in the brain, with disinhibitory chains onto EPG. So the model was corrected by splitting inhibition into two channels:

1. the structured Delta7 cosine surround, gated by the Delta7 gain.
2. a shallow ring-neuron term, $-rn \cdot \text{mean}(f(u))$, that scales with total bump mass and is *not* gated by Delta7.

With `rn_gain` 0.5 and `rn_depth` 0.5, the corrected model reproduces the real result across the whole sweep: width rises monotonically from 90° to 143° as Delta7 goes to zero, and the bump never dissolves.

There is a cost, and it is reported rather than hidden. The single-channel model scored 0.41 rad RMS on the tracking benchmark. The two-channel model scores 0.54 on the current benchmark (Chapter 12). Matching reality cost the model precision. I think the right reading of that is biological rather than a modelling failure: two inhibitory channels buy the fly robustness to losing one of them, at a price in tracking precision that a fly apparently does not need to pay for.

What the loop proved

Laid out as a sequence:

1. The LIF ensemble split into mechanism classes under perturbation.
2. The field model produced a quantitative curve the LIF ensemble could not express.
3. The two formalisms disagreed, and the disagreement was written into the artifact with a flag.
4. Published biology resolved it. The field model’s minority-matching outcome was right, but for an incomplete reason.

5. The model was corrected by adding the mechanism the data demanded, and the correction's cost was measured.

The refined prediction is now a quantitative width-versus-suppression curve, specific enough to check against imaging data point by point. And the LIF majority class, `d7_sculpts_sharp`, is disfavoured by both the corrected field model and the published experiment.

The general lesson is the one from Chapter 6. An ensemble cannot find a mechanism it does not contain. What it *can* do is refuse to reach a false consensus, and put its disagreement in a place where a second formalism and a paper can settle it. The hypothesis-lab artifact now carries a `cross_formalism` block for each ranked experiment, with the field-model outcome, the LIF class it maps to, and the agreement flag. When you open the file, you see the models arguing. That is what a hypothesis-testing pipeline should look like from the inside.

The PFN to $h\Delta B$ Circuit: A Wired Transform That Turns Out Not to Be One

The second operator through the lab is `phasor_vector_shift`. It matters for two reasons. Scientifically, it is the fan-shaped body’s coordinate transform, the step that turns a heading into the compass frame in which the fly computes travel direction. Methodologically, it is the test of whether the ensemble machinery is actually generic or secretly ring-shaped, because the geometry here is a line of columns, not a circle, and the observable is a centroid offset, not bump persistence.

What the wiring says

PFN neurons project to $h\Delta B$ with consistent column offsets: -3 for PFNd and $+2$ for PFNv in the structural histogram. Read naively, that is a fixed coordinate rotation compiled into the wiring. Drive column 6, and $h\Delta B$ should respond at column 3 through one arm and column 8 through the other.

The first version of the lab tested exactly that, and got a result that was directionally right and quantitatively wrong: realised offsets clustered at -1 to -2 , consistent in sign with the wiring but compressed in magnitude by dendritic integration, with the $h\Delta B$ response scaling sub-linearly with PFN drive.

The deeper workup started from the observation that the connectome contains three dynamical elements the static-offset picture ignores:

- PFN self-recurrence. 7,970 synapses within PFNd alone. The shift might be generated by dynamics rather than inherited from the projection.
- $h\Delta B \rightarrow$ PFN feedback. 61 edges. A recurrent loop the offset model treats as absent.
- $h\Delta B$ internal recurrence. Previously modelled as on or off, now swept continuously.

The ensemble

72 members over `pfndGain` $\times$ `pfnvGain` $\times$ `hdRecur` (0 to 2) $\times$ `hdTonic`, with each member also run under ablation of the three dynamical elements (`pfnRecur`, `hd2pfn`, and $h\Delta B$ recurrence set to zero). The probe drives PFN column 6 and measures where the $h\Delta B$ population responds, for each arm separately, and then sweeps the driven column to check whether the transform is rigid.

The result

Hypothesis classes: 13 `wired_shift`, 55 `passthrough`, 4 `silent`. The class counts are not the story. The mechanism attribution among the 13 shifted members is:

mechanism	members
needs_hdb_recur + hd2pfn_feedback	2
needs_hdb_recur + pfn_recur + hd2pfn_feedback	2
needs_hd2pfn_feedback	2
needs_pfn_recur	1
needs_hdb_recur + pfn_recur	1
needs_pfn_recur + hd2pfn_feedback	1
wired_only	0

Zero of thirteen shifted members produce the shift from the projection alone. Every one of them loses it when at least one of the recurrent elements is removed. Three further findings go with this:

1. The column sweep is never rigid. The realised offset changes with which column is driven. The transform is warped by position, not a clean linear shift.
2. PFNV rarely realises its +2. The PFNV arm has 20 cells against PFND's 40, and that asymmetry, invisible in the offset histogram, means the two phasor arms are not equivalent in the dynamics.
3. The generic runner adds arm coupling. When the circuit was rerun through the spec machinery of Chapter 13, some members' PFND shift required PFNV co-activity (`needs_pfnv_silenced` as a mechanism term), which the hand-written lab had not tested. The runner found 11 shifted, 56 passthrough and 5 silent at the same seed, again within a member or two of the lab.

The conclusion changed

I want to state plainly what happened here because it is the kind of thing that gets lost in a results table. The first-pass interpretation was “PFN→hΔB implements a wired phase shift, attenuated.” That is wrong. The correct statement is “the wiring provides a *bias*, and recurrence and feedback complete it into a shift.” Those are different mechanism classes, and they make different experimental predictions. A reader of the connectome alone would have published the wired offset as the circuit's function.

The ranked experiments

experiment	separation
amp_scaling	0.424
hd_recur_off	0.406
hd2pfn_off	0.316
measure_pfnd_offset	0.314
pfnv_silenced	0.312
sweep_rigidity	0.282

`amp_scaling` ranks first because the amplitude asymmetry between the arms is itself mechanism-dependent. The hΔB→PFN feedback loop, which was only added in the deeper workup, is already a top-three discriminator: removing it collapses the shift in the members that depend on it. So the experiment that would settle the mechanism, silencing the feedback while measuring the offset, is now known and ranked.

What this circuit teaches about the method

This is the clearest case in the book of structure overstating dynamics. The wiring says -3 . The dynamics say -3 is an upper bound that recurrence partially realises and position warps. The interesting number is not the offset. It is the gap between the wired offset and the realised one, because the gap is what tells you the transform is being computed rather than merely copied.

The Mushroom Body: When the Wiring Cannot Answer

The third circuit is the stress test the compiler was built for. `sparse_associative_memory` has no spatial geometry at all. There is no ring, no column axis, nothing to take a centroid of. The computation is pattern separation and associative readout, and the observables are similarity and gain. If the method only works where there is a coordinate system, this is where it shows.

The result is a negative, and I think it is the most important negative in the book.

The circuit

From Chapter 4, the wiring reads like a diagram of a random-projection memory:

- Expansion. 282 projection neurons reach 4,064 Kenyon cells, about $6\times$. Each KC samples a median 5 PNs, and the PN-to-KC sampling is near-random (correlation 0.023 against 0.013 shuffled).
- Normalisation. Two APL cells cover 100% of KCs and are driven by 100% of them, with 196k APL→KC synapses over 4,210 connections. That is one connection per KC, carrying about 48 synapses on average.
- Readout. 97 MBONs, a median 313 KCs each, 439k synapses.
- Teaching. About 340 dopaminergic neurons, 70% of whose MBON contacts are in the matching compartment.

The protocol

The memory geometry drives two overlapping KC odour patterns directly (200 cells, 50% shared) and measures:

- separation and expansion: the cosine similarity of the two MBON output vectors relative to the KC input overlap. Expansion above 1 means the readout decorrelates.
- compression: KC spike count at $2\times$ drive divided by count at $1\times$ drive. Divisive normalisation should make this well below 2.

The ensemble sweeps $kc2mb \in \{0.5, 1, 2\} \times aplGain \in \{0.5, 1, 2\} \times mbonTonic \in \{0, 4\}$ with $mbRecur = 1$, 18 members. The free axes come with their reasons: KC→MBON gain is plastic in vivo, APL feedback sets KC sparsity, and MBON→MBON recurrence exists (936 edges) but its role is unknown. Perturbations: silence APL, zero MBON recurrence.

The result

18 members: 9 silent, 9 collapsed. Zero gain_controlled. Zero linear_passthrough. The mechanism table is empty, because no member showed the mechanism.

The direct diagnostic explains why. Drive 400 KCs at increasing strength and count spikes:

drive	KC spikes	APL spikes (two cells)
60	4,163	13, 8
150	9,144	16, 11
300	14,338	16, 11

APL saturates at about 16 spikes in 200 ms while KC output scales $3.4\times$. The feedback cannot compress because APL's recruitment saturates, and two cells firing at 80 Hz through count-calibrated conductances cannot divide the output of thousands of KCs.

The ranked experiments confirm the picture from the other side. `baseline_gain_control` and `apl_silence` both score 0.529, and they sort the members identically: 9 silent and 9 uncontrolled, with or without APL. When silencing a population changes nothing, the population is not doing anything at this parameterization. Chapter 6 warned that a score near 0.5 in a two-class ensemble is a near-even split rather than a rich landscape, and this is that case.

What this means, and what it does not

There are three claims here and they need to be kept separate.

The wiring does not establish gain control. Whether the real APL divides KC output depends on the conductance per synapse, which the connectome does not carry. Synapse counts got the loop's existence, its coverage and its sign right. They could not get its strength. The lab reports this as a missing hypothesis class rather than as a tuned affirmative, which is the honest output.

The readout being linear is correct, not a failure. Expansion below 1 for 50% overlapping odours is what you expect when decorrelation happens upstream, at odour→KC, and the probe drives KCs directly and so skips that layer. The MBON transform is supposed to be near-linear. The memory's separation step lives in the expansion, and the probe did not exercise it.

The sign had to be checked. Early diagnostics found `nt: undefined` for APL. In this simulator, an undefined transmitter means the cell's edges contribute nothing, which would produce the same null result as above for a completely different reason. APL turned out to be correctly labelled: two cells, both GABAergic, and the loop is inhibitory as it should be. The weakness is quantitative, not a sign error. But had the check not been made, a bug and a finding would have been indistinguishable.

The same loop, in the whole brain

There is a result from the whole-brain model that looks like it contradicts this chapter, so let me explain why it does not.

In the full 165k-neuron LIF network (Chapter 14), driving Kenyon cells through their projection neurons and then silencing APL raises the fraction of active KCs from 0.15 to 0.72 and the mean KC rate from 0.87 to 5.08 Hz. APL is plainly doing something there.

Both results are true and they measure different things. The whole-brain probe asks whether APL keeps the KC code *sparse* under physiological PN input, where most KCs sit just below threshold and a small inhibitory conductance is enough to keep them there. The ensemble probe asks whether APL *divides* the output of KCs that are being driven hard and directly, where the same conductance is a small fraction of the drive. APL can be a thresholding sparsener without being a divisive normaliser. The catalogue predicted both. The wiring supports the first. It cannot decide the second.

The lesson

The connectome constrains a family of models, and the family's honest output sometimes includes "this cannot be determined from structure." The sparse-memory operator is detected with confidence 0.55, its wiring is canonical, its sparsening is realised, and its gain-control mechanism is underdetermined. Those facts coexist. And the failure is located precisely: "APL gain control needs per-synapse conductance values" is a more useful scientific output than either a false positive or silence, because it says which measurement would close the gap. If a future calibration gives APL enough conductance to matter, the `apl_silence` experiment stops being a coin flip and becomes the top discriminator, and the pipeline will already have said so.

The Engineering Benchmark: Is the Biological Estimator Any Good?

Once you have decompiled a circuit into an algorithm, you can ask a question that neuroscience usually cannot: is the algorithm competitive with what an engineer would have written? This chapter does that for heading estimation, and it tries hard not to cheat.

The task

Track heading $\theta(t)$ from noisy angular velocity plus intermittent landmark bearings, where 15% of the landmarks are corrupt (offset by π). Every navigation system solves a version of this. The ground truth is 60 seconds of Ornstein-Uhlenbeck angular velocity with saccade-like jumps, at 50 ms steps, with landmark observations every 2 s. All estimators see identical inputs. 60 trials, seeded, so the numbers below reproduce exactly.

The estimators

estimator	what it is	state
dead reckoning	integrate ω , ignore landmarks	1 float
complementary filter, $\alpha = 0.35$	integrate ω , correct toward landmark at fixed gain	1 float
complementary filter, $\alpha = 0.05$	same, at the correction strength matched to the ring model	1 float
Kalman, 1-D	constant-velocity Kalman filter on the wrapped angle, tuned	2 floats
Kalman with gate	same, with Mahalanobis outlier rejection of landmarks	2 floats
ring, scalar	the reduced ring attractor: PEN-drive integration plus a bounded landmark snap, as an	1 phase
ring field	8-wedge population code the full Amari field of Chapter 8, two-channel inhibition, advection velocity mode	256 units
ring field with cell noise	the same field with per-unit noise $\sigma = 0.15$	256 units

The matched-baseline rule is the one I would ask readers to check. The $\alpha = 0.05$ complementary filter exists so that the scalar ring model is compared with a classical filter applying the same effective correction strength. Without that row, a “win” for the ring could just be a gain choice.

Results

Numbers are RMS wrapped error in radians over 60 trials, with the standard deviation across trials, and the RMS error in the 5 seconds after a corrupt landmark:

estimator	RMS	$\pm$ sd	post-outlier
dead reckoning	0.661	0.285	0.589
complementary, $\alpha = 0.35$	0.724	0.196	1.176
Kalman, 1-D	1.159	0.263	2.044
Kalman with gate	0.546	0.418	0.784
complementary, $\alpha = 0.05$	0.622	0.211	0.630
ring, scalar	0.653	0.228	0.650
ring field	0.538	0.434	0.612
ring field with cell noise	0.521	0.391	0.588

These are the numbers for the corrected two-channel field model. The earlier single-channel version scored 0.41 on this task, and the version of this table in an earlier draft of the book showed 0.449. Matching the Delta7-block data in Chapter 9 cost about 0.1 rad. That cost is part of the result and I would rather report it than quote the older, better-looking number.

What the table says, in order of how much I believe it

At the scalar level there is nothing special. The reduced ring model (0.653) and the matched complementary filter (0.622) are a tie within noise. The scalar algorithm the attractor implements sits on the same trade-off curve as the textbook filters. If you reduce the fly’s heading circuit to a single phase variable, you have reinvented a complementary filter.

Clean tracking: the field ties the gated Kalman filter. 0.538 against 0.546, with standard deviations of 0.4 on both. I would not claim a win here and the table does not support one.

After a corrupt landmark, the field recovers better. 0.612 against 0.784 for the gated Kalman filter, and against 2.044 for the ungated one. The mechanism is structural. A corrupt landmark cannot simply shift a point estimate. It has to nucleate a competing bump on the field and win a spatial competition against the antipodal inhibition, which a single bad observation usually cannot do. The outlier robustness comes for free, with no gating logic and no threshold to tune. This is the claim I think the benchmark genuinely supports.

Per-cell noise costs nothing, and here slightly helps. With $\sigma = 0.15$ per-unit noise the field scores 0.521 rather than 0.538. The population representation averages independent noise across units, which is the thing a scalar estimator cannot do. I would treat the improvement as within noise and the absence of a penalty as the finding.

What this does and does not claim

This is not “the fly beats the Kalman filter.” The honest statement is narrower: in this regime, with corrupt landmarks, a spatial-competition estimator is a competitive mechanism class, and its robustness is architectural rather than tuned.

The caveats are real. The benchmark is tuned to one noise regime. The excitatory gain of the field is a free parameter, because the wiring shows $EPG \leftrightarrow EPG$ and PEG recurrence exist but not their strength. Real PEN neurons are graded, not the idealised advection the best-scoring mode uses, and Chapter 8 showed that the more realistic position-coded mechanism sits about 50% above the advection bound. Put those together and the biological circuit, modelled faithfully, probably lands closer to the gated Kalman filter on clean tracking while keeping its advantage after outliers. That is still an interesting engineering result. It is just not a dramatic one, and the book should not make it sound like one.

The Compiler: Turning a Detected Operator into an Executed Ensemble

The problem with three hand-written labs

The first version of the pipeline had a weakness that the circuits themselves exposed. The ring lab, the phasor lab and the memory lab were three separate scripts, each with its circuit's populations, observables and perturbations written into the code. That is “we ran three analyses.” It is not “the compiler compiles.” If adding a fourth circuit means writing a fourth script, then the operator catalogue of Chapter 5 is a list of ideas rather than an input to anything.

The fix is a middle layer: a declarative spec per operator, and one generic runner that executes any spec.

Declarative specs

A spec generator reads the operator catalogue and emits, per detected operator, the circuit-specific content as data. The mushroom-body spec, abbreviated:

```
{
  "geometry": "memory",
  "operator": "sparse_associative_memory",
  "populations": {"kc": "KC*", "apl": "APL*", "mbon": "MBON*", "dan": ["PPL*", "PAM*"]},
  "roles": {"input": "kc", "output": "mbon", "control": "apl"},
  "odor_size": 200, "odor_overlap": 0.5,
  "edge_params": [
    {"pre": "KC*", "post": "MBON*", "param": "kc2mb", "why": "KC->MBON readout gain (44k e"},
    {"pre": "APL*", "post": "KC*", "param": "aplGain", "why": "APL feedback inhibition sets"},
    {"pre": "MBON*", "post": "MBON*", "param": "mbRecur", "why": "MBON->MBON recurrence (936 e"},
  ],
  "tonics": [{"pop": "mbon", "param": "mbonTonic"}],
  "grid": {"kc2mb": [0.5, 1, 2], "aplGain": [0.5, 1, 2], "mbonTonic": [0, 4], "mbRecur": [1]},
  "perturbations": [
    {"name": "apl_silence", "silence": "apl", "why": "does separation survive without feedback"},
    {"name": "mb_recur_off", "param": "mbRecur", "set": 0}],
  "hypotheses": ["gain_controlled", "linear_passthrough", "collapsed", "silent"],
  "provenance": {"signature": "...", "evidence": [...]}
}
```

Populations resolve by type prefix, so KC* is every Kenyon-cell subtype. Roles name the functional slots.

Each free axis carries a why: the reason the wiring does not fix it. Each perturbation carries a why too. And the provenance block copies over the evidence items that justified the detection in the first place, so the artifact can be traced from wiring to claim without opening another file.

The ring spec’s axes, for comparison, are EPG→EPG recurrence (“not fixed by synapse counts”), Delta7→EPG gain (“glutamatergic via GluCl α , effective strength uncertain”), PEN→EPG gain, and EPG tonic bias (“resting excitability unknown”). Its d7_silence perturbation is annotated “top-ranked discriminator in the LIF ensemble.” The spec is where the reasoning lives, in a form a program can consume.

The generic runner

One script executes any spec against the connectome. It handles everything that is circuit-independent:

- population resolution, exact and by prefix.
- the ensemble grid over the free axes.
- the probe, perturb, classify and rank loop.
- mechanism attribution by ablating each dynamical element in each member.
- the JSON artifact.

What is *not* generic is the observable geometry: how you measure whether the computation happened. That lives in a plugin per geometry class:

geometry	operator	observable
ring	ring_attractor	bump phase, width and angular velocity on the circle
linear	phasor_vector_shift	centroid offsets on a column axis, arm by arm
memory	sparse_associative_memory	pattern separation and gain compression

A fourth operator will need a fourth plugin if its geometry is new. `motion_correlator`, for instance, needs synaptic delays and direction-selective pairing, which none of the three plugins expresses. I actually think that is a feature: “what geometry class is this operator” is itself a meaningful classification, and being forced to answer it is better than pretending a ring observable applies to a filter bank.

Does the compiled version reproduce the hand-written one?

Yes, to the precision one should expect:

circuit	hand-written lab	spec through the runner
ring, seed 42	30 silent / 12 attractor / 6 filter, best tonic_sweep 0.414	30 / 11 / 7, best tonic_sweep 0.393, same top-three set
phasor, seed 42	13 shifted / 55 passthrough / 4 silent, best amp_scaling 0.424	11 / 56 / 5, best amp_scaling 0.461, plus the arm-coupling mechanism the lab did not test

circuit	hand-written lab	spec through the runner
memory	(first run through the runner)	9 silent / 9 collapsed, 0 gain-controlled

The differences are one or two members crossing a class boundary, which is the size of discrepancy you get from two code paths sharing a graph but not a random-number stream or an integration order. If the two had matched exactly I would have suspected shared code rather than independent confirmation.

What “compiled” means now

Adding the mushroom-body operator required one spec entry, about 30 lines of data, and one geometry plugin, about 60 lines. The runner did the rest. So the compiler claim is real, narrowly:

detected operator + measured structure → declarative spec → executed ensemble → ranked experiment table, with no circuit-specific code in the loop.

The one remaining hand-authored step is the spec itself. A human still chooses the free axes, using knowledge of which parameters the wiring leaves open. Automating that is mechanical, and it is the first item in Chapter 15’s list: every unmeasured edge class becomes an axis, every substrate population becomes a perturbation.

The artifact schema

Every run writes the same shape: `seed`, `summary` (ensemble size, class counts, best experiment and its separation), `members` (parameters, baseline observables, perturbation outcomes and mechanism label per member), `ranked_experiments`, and where a field model exists, a `cross_formalism` block. The web viewer and the report generator consume that one schema, so a new circuit’s results are browsable the moment they exist.

Running the Whole Brain in a Body

Everything so far has been about circuits of a few hundred cells. This chapter is about what happens when you take the other route and run all 165,122 neurons at once, in a physics-simulated body, in a browser. I think of it as the complement to the compiler: the compiler tells you *why* a circuit works or fails, the whole-brain run tells you *whether the wiring as a whole supports behaviour*, and each is uninterpretable without the other.

The result, in one sentence: the raw connectome, with nine fitted global parameters, reproduces a set of published reflexes and sensorimotor mappings, and does not by itself produce coordinated walking, spontaneous behaviour, or the state-dependent drive that real flies have. Everything in the second list had to be supplied by code outside the graph, and the interesting content is the exact boundary between the lists.

The setup

The brain is the conductance-based LIF model of Chapter 6, over all 10.5 million connections, stepped at 0.5 ms in WebAssembly or WebGPU. The body is flybody (Vaxenburg et al. 2024): 67 bodies, 102 joints, 78 actuators, adhesive tarsal claws, total mass 0.98 mg, stepped by MuJoCo at 0.2 ms. The eye is flyvis (Lappalainen et al. 2024), a trained optic-lobe model on a 721-column hexagonal lattice per eye, whose outputs drive the matching 62,000 male-CNS optic-lobe neurons by cell type and retinotopic column. Taste, smell, touch, proprioception, wind and heat enter as Poisson spike trains on 7,745 sensory neurons in 151 channels, at rates taken from physiology. The motor side reads firing rates from identified descending neurons and 439 motor neurons mapped to 170 muscle groups.

Throughput is about $0.2\times$ real time per fly with vision, five flies at once on a 12-core laptop. Two implementation details matter here because they are model claims, not engineering trivia. First, an event-driven update was tried and gained almost nothing, because about 120,000 neurons are subthreshold-active at any step. The network is not sparse in time. Second, the WebAssembly and WebGPU kernels were checked against each other on 1,000 identically driven steps: 10,298 against 9,580 sampled spikes, which is within-seed agreement for a stochastic model, and a deterministic bias-driven chain gave spike counts 16, 8, 6, 5 exactly on both.

What had to be added to the graph before it worked at all

Each of these is a place where the connectome, as a file, is missing something a neuron has:

addition	why
conductance-based synapses, E_{exc} 0 mV, E_{inh} fitted to -76 mV PSP scaled by $(\text{volume} / \text{regional median})^{-0.61}$	current-based synapses let any neuron with 150 synapses fire any target larger neurons have lower input resistance (Pugliese et al. 2025)
graded signs for 3,602 neurons without a consensus transmitter sensory neurons ignore central input	guessing “excitatory” spread odour to every glomerulus their spikes start at the receptor; central synapses on their terminals cannot fire them
raised Kenyon-cell threshold (+10.9 mV) lamina resting bias (+5.3 mV)	KCs need coincident input from several PNs L1 to L5 are graded cells with a depolarised rest, so histaminergic photoreceptor input can modulate them
giant fibre to TTMn electrical synapse	absent from the chemical connectome, present in the animal, and the escape jump does not happen without it
octopamine as a slow modulator	octopaminergic neurons have no fast synapses; their release lowers targets’ thresholds over seconds

Three things were tried and dropped, and I list them because the dropped ones are as informative as the kept ones. A firing-rate model of the whole CNS after Pugliese et al. ignited to about 25,000 active neurons and gave slow waves rather than a stepping rhythm. Spike-frequency adaptation and short-term depression stabilised activity but killed the sugar-to-proboscis pathway, which is why the fitted adaptation is zero. Treating unknown transmitters as excitatory flooded the antennal lobe.

Calibration against published physiology

Nine global parameters were fitted by cross-entropy search (20 generations of 24 candidates, 12 evaluators in parallel, 2 to 4 seconds per evaluation) against a benchmark suite of published results:

target	source	model
labellar sugar drives the proboscis motor neuron MN9	Shiu et al. 2024	59 Hz
bitter silences MN9	Shiu et al. 2024	0 Hz
bitter vetoes sugar	Shiu et al. 2024	0 Hz
front-leg sugar drives MN9 (tarsal reflex)		34 Hz
leg sugar suppresses the walking drive		$9.2 \rightarrow 4.6$ Hz
leg sugar does not drive backward walking		0.6 Hz
Kenyon-cell sparseness	Turner et al. 2008	7.4%
looming drives the takeoff neurons over self-motion	von Reyn 2014, Namiki 2018	33 vs 2 Hz

target	source	model
no giant-fibre spikes during self-motion		pass
BDN2 activates leg muscle groups	Pugliese et al. 2025	pass
odour specificity in KCs and PNs		failed every configuration

The overall score is 0.746 on the best run, 0.725 mean of six. An earlier fit without neuromodulation scored 0.767 on one run and 0.674 mean of six, which is a useful reminder that single-run scores on a stochastic model mean very little.

The odour-specificity failure is the calibration’s cleanest negative. Chapter 4 already gave the structural reason: cholinergic antennal-lobe local neurons make about 200,000 synapses onto projection neurons, and no global parameter setting keeps the channels apart downstream. This is a circuit whose function is not recoverable from wiring plus nine knobs.

Two benchmark-design bugs were found along the way and belong here because they generalise. An objective that rewarded odour specificity was satisfied, briefly, by a configuration in which no Kenyon cell fired. And reused WebAssembly memory carried spike counts between evaluations until every state array was reset. Both are the kind of thing that produces a confident wrong number.

What the raw wiring does when you let it move

The most informative measurements are the ones made *before* any behavioural module was added.

With the body held still, the forward-walking descending neurons were silent. Once the body was allowed to move, footfall touch bursts drove those same neurons to 10 to 13 Hz and the fly ran at full speed without stopping. Masking tarsal touch brought them back to about 3 Hz. The same bursts made the steering neurons alternate, and steering was pinned at its limit 20 to 50% of the time.

So the connectome-derived network is reactive. Its locomotor output is a function of its own refference, not of any internal state. Insects solve this with presynaptic inhibition of afferents during self-generated steps, and the model does the same: footfall bursts are scaled by $1 - 0.85 \times \text{stepping amplitude}$. That 0.85 is a parameter the connectome cannot supply and behaviour demands.

A second example of the same kind: constant contact with food kept driving the walking neurons, and the fly could not stop on it. Tactile bristles are rapidly adapting in the animal, and once the model’s were made so, it could.

A third, which I find the most striking: the reconstructed wiring is not left-right symmetric. The right DNa02 and P9 receive more tonic excitation than the left, and the left P9 receives about twice the inhibitory conductance of the right. Without a slow adaptation (4 s time constant) on the steering readout, the fly circled right. A bilateral-symmetry check should probably be a standard test for any connectome model.

What the connectome supports, and what it does not

Supported by the wiring, read out from identified descending neurons:

- Forward walking (DN_g100/BDN₂, DN_g97/oDN₁, DN_p09/P₉), backward walking (MDN), steering (DN_a02, DN_a01), grooming (DN_g07, DN_g08, DN_g12), escape (the giant fibre, DN_p01) and takeoff (DN_p02, DN_p04), courtship (pIP₁₀, DN_p13).
- A sensory screen over which senses drive which commands. Dimming or looming drives the giant fibre and takeoff neurons. Wind on the antennae and hind-leg touch drive the backward-walking neuron. Many odours drive the steering neurons. The forward-walking neurons are driven mostly through vision.
- Courtship detection. Given a pheromone plume and a small-object visual signal at LC₁₀, the connectome propagates both to pIP₁₀ and DN_p13, which roughly double their firing near another fly. Only the LC₁₀ input is injected. The decision variable is computed by the wiring.
- Sparsening in the mushroom body, at 7.4% active Kenyon cells.

Not supported, supplied from outside the graph:

- Walking itself. No connectome-only model produces coordinated walking from the whole nerve cord. Pugliese et al. found rhythm in a front-leg subnetwork for about 3% of descending neurons. The full-connectome motor mode, where every leg muscle is driven by its own motor neurons through the raw VNC wiring, cannot hold posture. That is reported as the honest result, and a stepping-pattern generator fitted to 100 real walking trajectories (median 9.5 Hz step, 1.7 cm/s, 12° RMS joint-angle error) sits between the descending neurons and the legs instead.
- Spontaneous behaviour. Bout structure is not in the wiring. Walk bouts (lognormal, median 2.2 s), pauses (1.4 s), grooming in 20% of pauses, saccades at 0.7 per second walking with 65% alternation, post-feeding local search: all of this comes from an endogenous module with sources in the ethology literature (Maye 2007, Seeds 2014, Geurten 2014, Dethier 1957). It never writes actuator commands. It delivers excitatory conductance to identified descending neurons, so every command still passes through the connectome. Conductance rather than current, because the embodied network holds those neurons in a high-conductance state with inhibition about three times the leak, and 9 mV of injected current moved them by about 2 mV.
- Escape gating. Without it the fly jumped repeatedly into walls, because a wall you walk into looms on the eye, and so do your own grooming legs. Real flies separate these by touch, matched optic flow and efference copy.
- The escape jump and righting, as explicit motor programs. The jump lands upright in 24 of 24 trials. Leg-only righting never works because the legs cannot reach the floor from the back, and the wing-assisted program rights the fly in 4 of 6 inverted starts.

Weakly supported, and reported as such:

- Looming escape is intermittent: 2 of 10 test looms produced an escape. The takeoff neurons reach their 70 Hz trigger during self-motion and grooming, and real looms reach only 70 to 100 Hz, so the discriminator has essentially no margin. Pathway-specific fitting of the LC₄, LPLC₂, giant fibre, DN_p02 and DN_p04 chain is the top open item.
- Turning away from touched obstacles and from heat. The connectome responds to both senses but steers away only weakly.
- Feeding completes only with an endogenous stop and hunger-gated MN₉ drive. The model's own sugar-stop pathway is too weak on its own.

Neuromodulation: two negatives worth more than a positive

The most instructive part of the whole-brain work is the octopamine chain, because it was built faithfully and did not do what the literature says it should.

The chain runs as follows. Sugar level sets AKH, which is secreted outside the CNS and so is not in the connectome, and insulin, from 16 insulin-producing cells that are. AKH excites and insulin inhibits 17 octopaminergic cells (Yu et al. 2016). 37 octopamine neurons release in proportion to their firing, and release lowers the spike threshold of all 12,851 of their targets by up to 2 mV. Receptor types per cell are not in the connectome, so all targets are treated alike.

Three model changes were needed to make this work at all. Octopamine’s fast synapses were removed from the LIF graph, since the cells have none. A slow afterhyperpolarisation (threshold up 0.5 mV per Hz of the last 2 s) was added, because without it the modulatory neurons switched between silence and 100 Hz for small changes in input. And their resting thresholds were calibrated so a fed fly’s octopamine cells sit at 2 Hz instead of the 100 to 200 Hz the raw wiring produced.

Then the genotype experiment, matching Yang et al. 2015 and Lee and Park 2004:

genotype	fraction moving, fed	starved	octopamine Hz, fed → starved
wild type	0.55	0.73	3.0 → 13.4
Tβh null (no octopamine synthesis)	0.51	0.56	1.8 → 10.7
AKHR null	0.64	0.65	2.5 → 7.4
direct (modulation on, bout rules blind to it)	0.60	0.57	3.0 → 13.2

The wild type and the two knockouts match the published phenotypes. The fourth row is the negative. When octopamine acts on the connectome but the bout rules are not allowed to read arousal, starved flies walk *less* (1.1 versus 1.3 cm/s). The wiring gives a reason: among the descending neurons the octopamine cells contact, the steering and backward-walking neurons receive more octopamine synapses than the forward-walking ones. A correctly wired, correctly calibrated modulatory broadcast produces the wrong sign of behavioural change, because the target distribution favours turning over forward drive.

The second negative concerns the loom. Removing octopamine’s fast synapses cost the giant fibre its response to a tethered loom (0.8 spikes → 0) and dropped the takeoff neurons from 45 to 25 Hz, apparently because the optic-lobe octopamine cells had been adding visual gain through synapses that should have been modulatory. Refitting recovered the takeoff neurons (about 30 Hz against 2 Hz for self-motion) but not the giant fibre. Implementing the real mechanism, a locomotor corollary discharge that raises optic-lobe octopamine release 2 to 3× during walking (Suver et al. 2012), also did not bring it back. So the weak link in the loom chain is upstream of, or parallel to, the octopamine gain, and the model can say that much precisely.

What this chapter adds to the compiler's picture

The circuit-level chapters found that structure overstates dynamics in three circuits. The whole-brain run finds the same thing at the level of behaviour, and adds a list of what fills the gap: nine fitted scalars, one missing electrical synapse, a refference gain, an adaptation time constant, a bout-statistics module, and a stepping generator. Every item on that list is a measurement the connectome does not contain and a behaviour depends on. That is not an argument against connectomes. It is the most specific statement I know of about what to measure next.

Synthesis: What We Have Got

The result that keeps recurring

Across three circuits, three observable geometries, two model formalisms and one whole-brain embodied run, the same thing happens:

Structure systematically overstates what dynamics delivers.

- The Delta7 kernel fits the wiring at R^2 0.99 and the realised inhibition at R^2 0.56. The attractor it belongs to exists only with a tonic bias of 3 to 7 mV that the wiring says nothing about.
- The PFN $\rightarrow$ h Δ B wiring says “shift by 3 columns.” The dynamics realise a smaller, warped shift, and only when recurrence and feedback participate. Zero of thirteen shifted members do it from the projection alone.
- The mushroom-body wiring shows the canonical APL normalisation loop with 100% coverage. At count-calibrated weights the loop sparsens but cannot divide.
- The whole nerve cord, wired as reconstructed, cannot hold posture. The escape discriminator has no margin. The octopamine broadcast pushes behaviour the wrong way.

I want to argue that this is a calibration signal, not a counsel of despair. The gap between what structure advertises and what dynamics delivers is precisely the quantity that tells you where to look next: the missing tonic bias, the missing recurrence gain, the missing per-synapse conductance, the missing electrical synapse, the missing refference gain. Each gap is a specific, locatable demand for data. And each was found by a method that could have hidden it with a tuned parameter and chose not to.

What the machinery can do now

In order of increasing strength:

1. Structural analysis that transfers across species. The same passes run on the worm and the fly and produce comparable numbers, including a found-not-annotated identification of each animal’s dominant dynamical mode.
2. Ensemble fitting. Wire-constrained model families, classified by behaviour, with the free axes and their reasons written into a spec.
3. Mechanism discrimination. Perturbations ranked by how completely they split the surviving hypotheses, stable across seeds.
4. Cross-formalism checking. Spiking and field models compared on the same perturbation, with disagreement recorded in the artifact.
5. Correction against withheld data. Published biology used to fix a model’s mechanism, not just to validate it, with the cost of the fix measured.

6. Negative results as a first-class output. “The wiring underdetermines this mechanism” reported as a missing hypothesis class, with the missing measurement named.
7. Embodied validation. The whole graph run in a body against published behaviour, with a written boundary between what the wiring supports and what was supplied.

What the connectome cannot determine

The failures are catalogued as carefully as the successes, and they come in two kinds.

Missing from the data:

- Per-synapse conductance. Synapse counts are not PSPs, and the counts are nearly incompressible given everything else in the graph, so nothing else predicts them. Chapter 11 is the case study.
- Gap junctions. Absent from the reconstruction, present in the animal, decisive for the escape jump.
- Neuromodulatory receptors. Which cells carry which receptor for octopamine, dopamine or serotonin is not in the file, so a modulatory broadcast has to treat all 12,851 targets alike.
- Time constants, delays, adaptation, and their heterogeneity. The models carry one value each where biology carries a distribution.
- Connections under three synapses, and any synapse the reconstruction missed.

Missing from the method, so far:

- Mechanisms the ensemble does not contain. Chapter 9 shows the failure mode and the fix. It does not show a general way to know when a mechanism is missing.
- Circuits with no geometry plugin. The motion correlator needs delays. The antennal lobe needs a plugin that expresses glomerular identity.
- Behaviour that lives in dynamics the LIF model does not have. Bout structure, central pattern generation, and state-dependent gain are all things the animal does with cells the connectome contains, and the model does not reproduce them from the wiring. Whether that is a missing parameter or a missing mechanism is not yet known.

The right framing is that these are the boundaries of the data and the current method, and that the pipeline’s job is to *locate* them precisely. “APL gain control needs conductance values” and “the loom chain’s weak link is upstream of octopamine gain” are more useful outputs than either a false positive or silence.

Open problems, in the order I would do them

1. Derive the spec axes automatically. Every unmeasured edge class becomes a free parameter, every substrate population becomes a perturbation. This removes the last hand-authored step in the compiler.
2. A fourth operator with a new geometry. The T4/T5 motion correlator needs synaptic delays. Antennal-lobe divisive normalisation re-asks the APL question on a circuit with hundreds of interneurons instead of two.
3. Pathway-specific fitting of the escape chain. LC4, LPLC2, giant fibre, DNp02 and DNp04 against loom data, the way flyvis was fitted for the optic lobe. It is the clearest weak link in the embodied model.

4. Quantitative withheld validation beyond the Delta7 curve. The width-versus- suppression curve is specific enough to check against imaging data point by point.
5. Fold cross-formalism agreement into the experiment ranking. Experiment scores currently weight only within-LIF disagreement. A disagreement that survives both formalisms should rank higher.
6. Close the PFN $\rightarrow$ h Δ B loop. The feedback perturbation is a top-three discriminator and the experiment is known.
7. A third species. The MICrONS cortical volume has connectivity plus measured activity, which would make every detected operator falsifiable against real responses rather than against the literature.
8. Bout structure and dopamine gating from circuits rather than rules. The whole-brain model's largest remaining piece of non-connectome code.

The claim, restated

The connectome is a compiled artifact: the output of a developmental program that built a machine whose description is a wiring diagram. This book documents a decompiler. It reads the diagram, proposes the computations, builds the model families the diagram permits, and identifies the experiments that decide between them. It also runs the whole diagram in a body to see which behaviours fall out.

It works where it should: the ring attractor is recovered, its mechanism is derived, and its model is corrected against real data. It fails where it should: the mushroom body's missing parameter and the nerve cord's missing rhythm are located rather than papered over. And it reports the difference throughout, which is the only property of a method like this that I would insist on.

Appendix A: Reproducing the Numbers

Every number in this book comes from a committed script and, in most cases, a committed artifact. The random-number generators are seeded, so rerunning a stage reproduces the cited table. Where a stage was rerun for this draft (the benchmark in Chapter 12) the new numbers replaced the old ones and the text says so.

Pipeline order

```
# IR and generic structure (Chapters 2 to 4)
python3 scripts/algo_structures.py      # fly structural report, ~3 min -> public/data/algo_stru
python3 scripts/connectome_ir.py        # fly + worm -> canonical IR
python3 scripts/algo_ir.py              # IR -> generic cross-species report -> public/data/ir_g
python3 scripts/check_structures.py     # fly-through-IR reproduces algo_structures.json
node scripts/validate_dynamics.mjs     # structure-vs-LIF checks -> public/data/dynamics_valida

# Operator detection (Chapter 5)
python3 scripts/operators.py            # structure report -> public/data/operators.json

# Ensembles (Chapters 6, 7, 10, 11)
python3 scripts/ensemble_spec.py        # operators -> public/data/ensemble_specs.json
node scripts/run_ensemble.mjs ring_attractor
node scripts/run_ensemble.mjs phasor_vector_shift
node scripts/run_ensemble.mjs sparse_associative_memory
node scripts/hypothesis_lab.mjs 42      # the hand-written ring lab, seed 42
node scripts/phasor_lab.mjs 42        # the hand-written phasor lab

# Field model and perturbations (Chapters 8, 9)
python3 scripts/ring_pde.py             # self-check: bump, integration, landmark remap
python3 scripts/perturb_pde.py         # -> public/data/perturb_pde.json

# Benchmark (Chapter 12), ~40 s
python3 scripts/bench_heading.py

# Whole-brain model (Chapter 14)
node scripts/calib_eval.mjs             # one parameter set against the benchmark suite
node scripts/calib_search.mjs          # cross-entropy search
node scripts/behavior_report.mjs       # closed-loop scenarios
```

```

node scripts/sensory_screen.mjs      # which senses drive which commands
node scripts/starvation.mjs          # the genotype table
node scripts/neuromod_calib.mjs      # modulatory-neuron resting thresholds

```

Key artifacts

artifact	content
public/data/algo_structures.json	full structural report for the fly
public/data/ir_generic.json	cross-species IR comparison
public/data/dynamics_validation.json	Delta7 kernel, PEN push field and APL sparsening in the LIF model
public/data/operators.json	the operator catalogue
public/data/ensemble_specs.json	the declarative specs
public/data/hypothesis_lab.json	ring ensemble, ranked experiments, cross-formalism block
public/data/ring_attractor_lab.json	the same circuit through the generic runner
public/data/phasor_lab.json, phasor_vector_shift_lab.json	PFN→hΔB ensemble, lab and runner
public/data/sparse_associative_memory_lab.json	memory ensemble
public/data/perturb_pde.json	field-model perturbation curves
public/data/brain_params.json	the fitted whole-brain parameters

Notation

- Ensemble *members* are parameter points. *Classes* are behaviour labels assigned by the probe. *Mechanisms* are the perturbation-outcome subclasses within a class.
- Separation scores are the fraction of member pairs an experiment places in different classes, in $[0, 1]$. A score near 0.5 in a two-class ensemble is a near-even split.
- Bump widths are full width at half maximum in degrees. Benchmark scores are RMS wrapped error in radians.
- *Wired* means read from synapse counts. *Realised* means measured in simulation.
- Column offsets in the central complex are in protocerebral-bridge glomeruli (45° each, 8 per side). A PEN shift of 1.5 columns is one wedge of 22.5° .

Appendix B: References and Sources

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